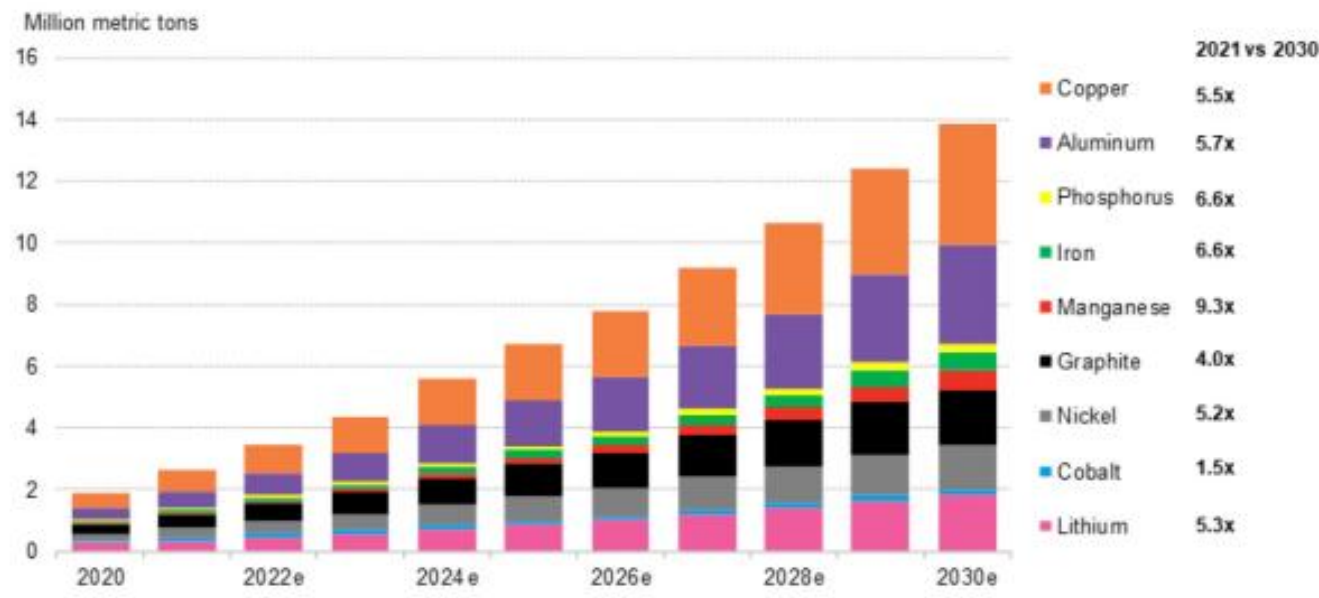


Deep-Sea Minerals New Challenges for Geophysics

Ketil Hokstad, Equinor Research Centre Trondheim
Lofotenseminaret i Geofysikk, 2. March 2022

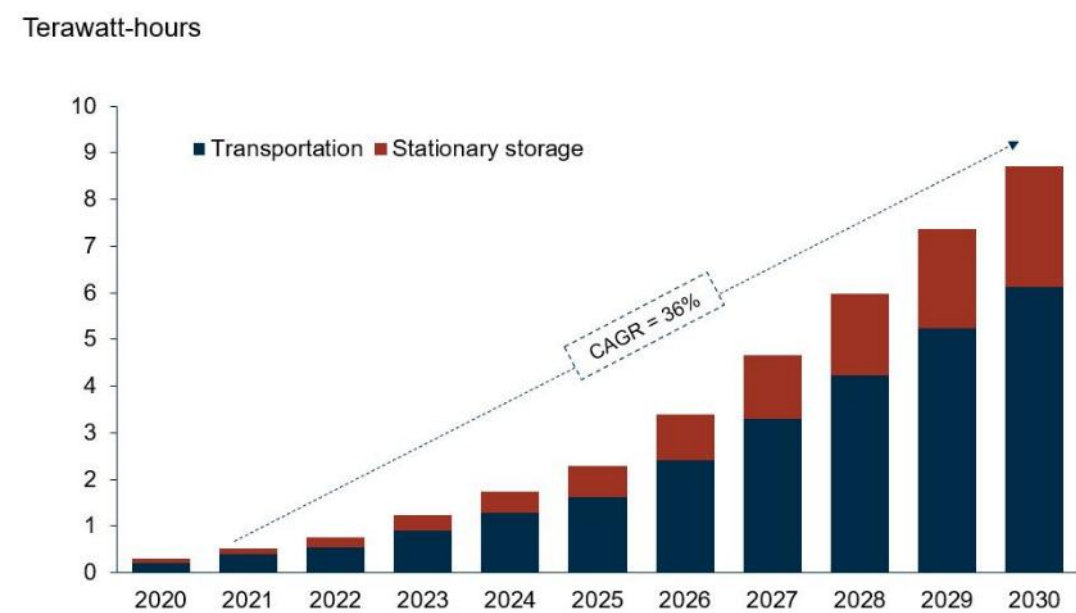
The green shift – need for metals

Metal demands for Li-ion batteries



Source: Bloomberg

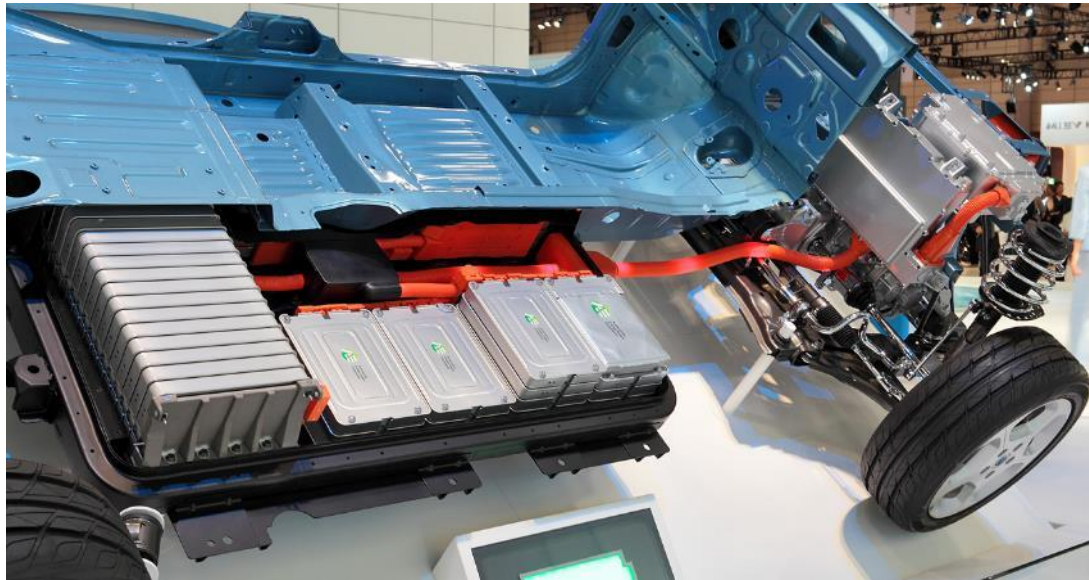
Battery-demand forecast by sector



Source: Rystad Energy

75 kWh EV batteri (NMC811)

- 80 kg Cu (25 kg for ICE)
- 56 kg Ni
- 6.6 kg Mn
- 7.1 kg Co
- ~10 kg Li



Picture from Wikipedia

Offshore windfarm export cable:

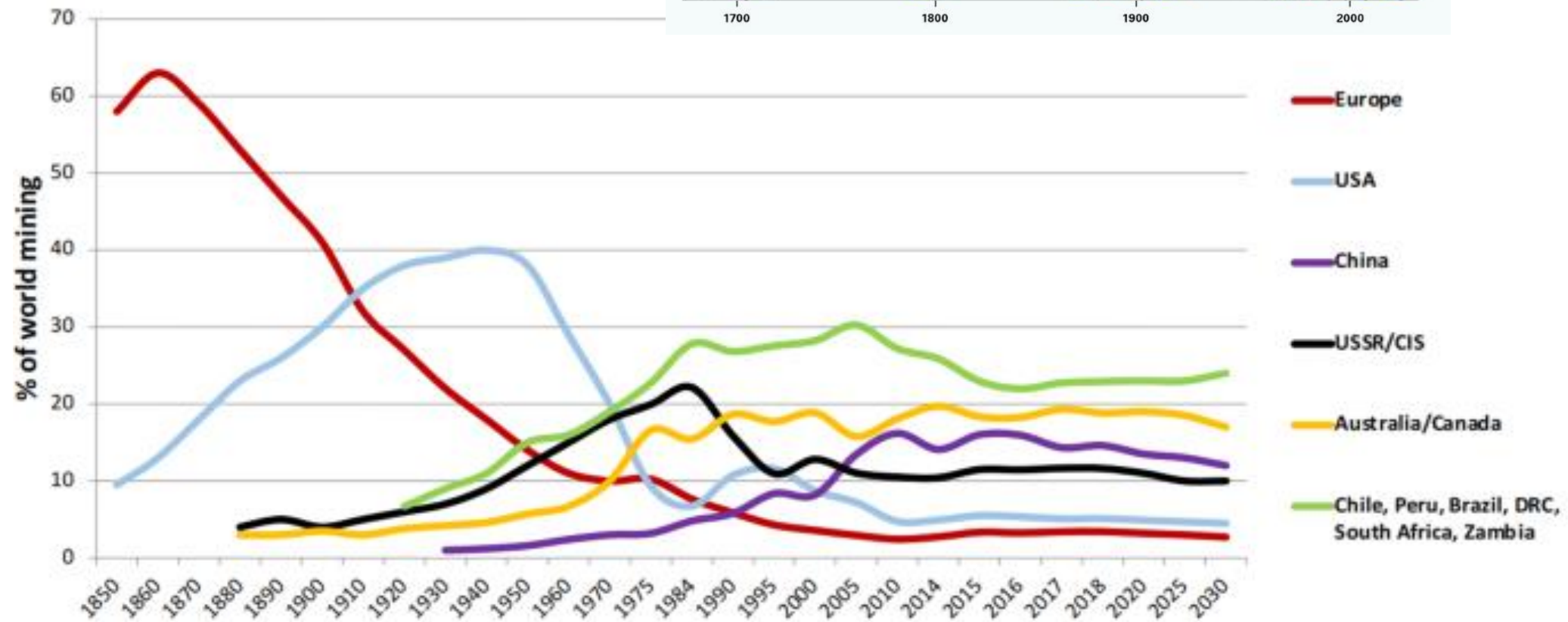
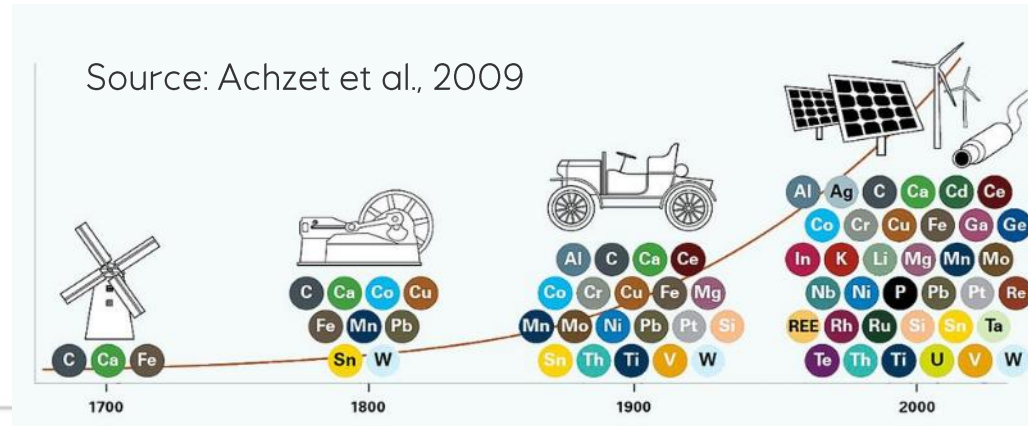
- ~50 kg Cu per meter



Picture from nexans.no

World of mining 1850 - present

Metals and minerals



Source: RMG Consulting 2021

Three types of deep-sea mineral resources

Seabed massive sulphides

Polymetallic nodules

Cobalt-rich crusts

Seamount.

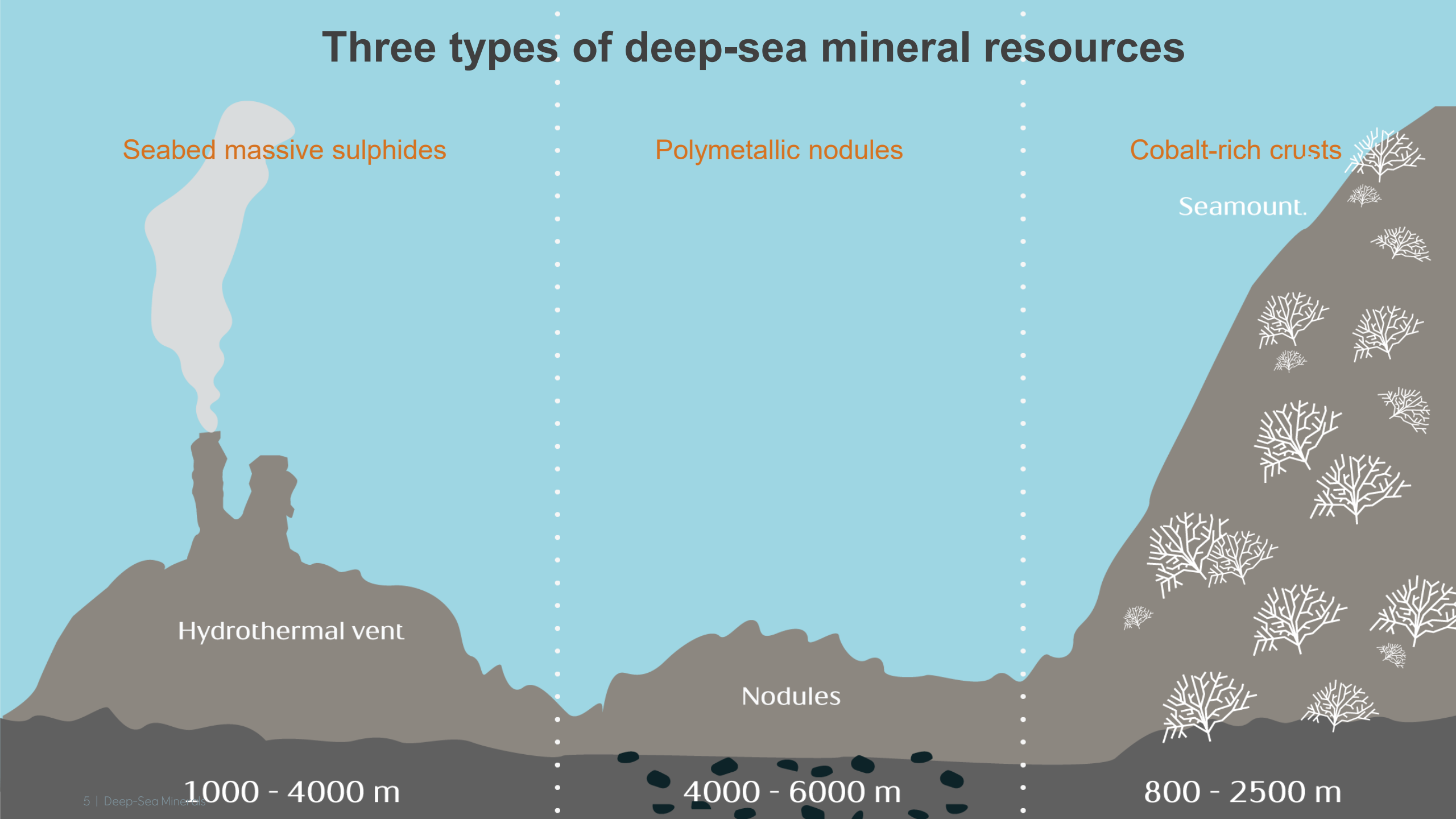
Hydrothermal vent

Nodules

1000 - 4000 m

4000 - 6000 m

800 - 2500 m



Exploration play concepts

Petroleum

«Petroleum system analysis»

- Source
- Reservoir
- Trap/Seal



Geothermal

«Play fairway analysis»

- Heat source
- Recharge
- Producibility



Seabed massive sulfides

«Mineral systems analysis»

- Metal source
- Fluid pathway
- Accumulation and preservation



Polymetallic nodules

«Mineral systems analysis»

- Metal source (cation&anion)
- Metal transport
- Favorable conditions for nodule growth



If one play element fails, the play fails

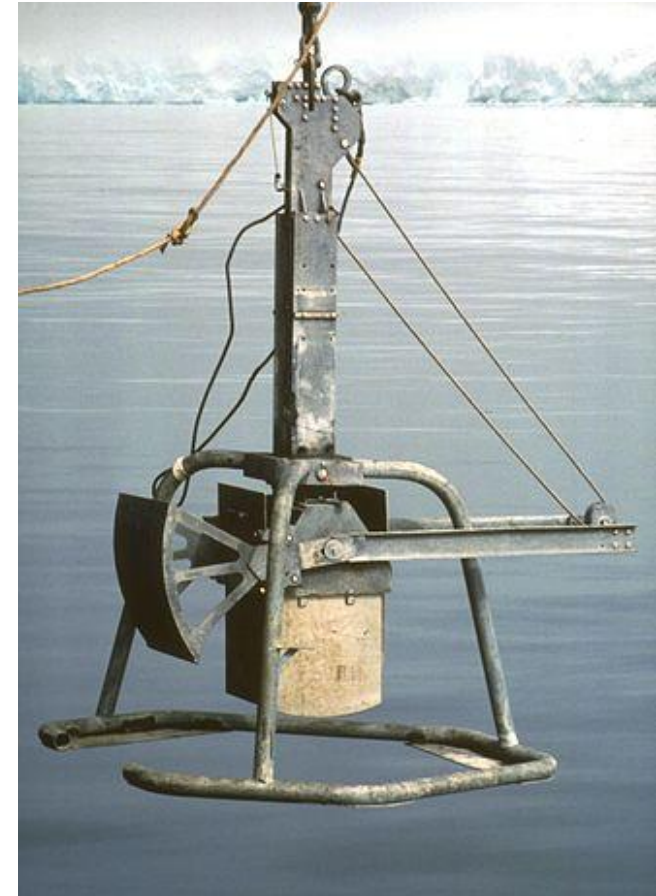
Polymetallic nodules "battery in a rock"

Cathode material for Li-ion batteries



Patania II nodule harvester (GSR)

Source: mining-technology.com



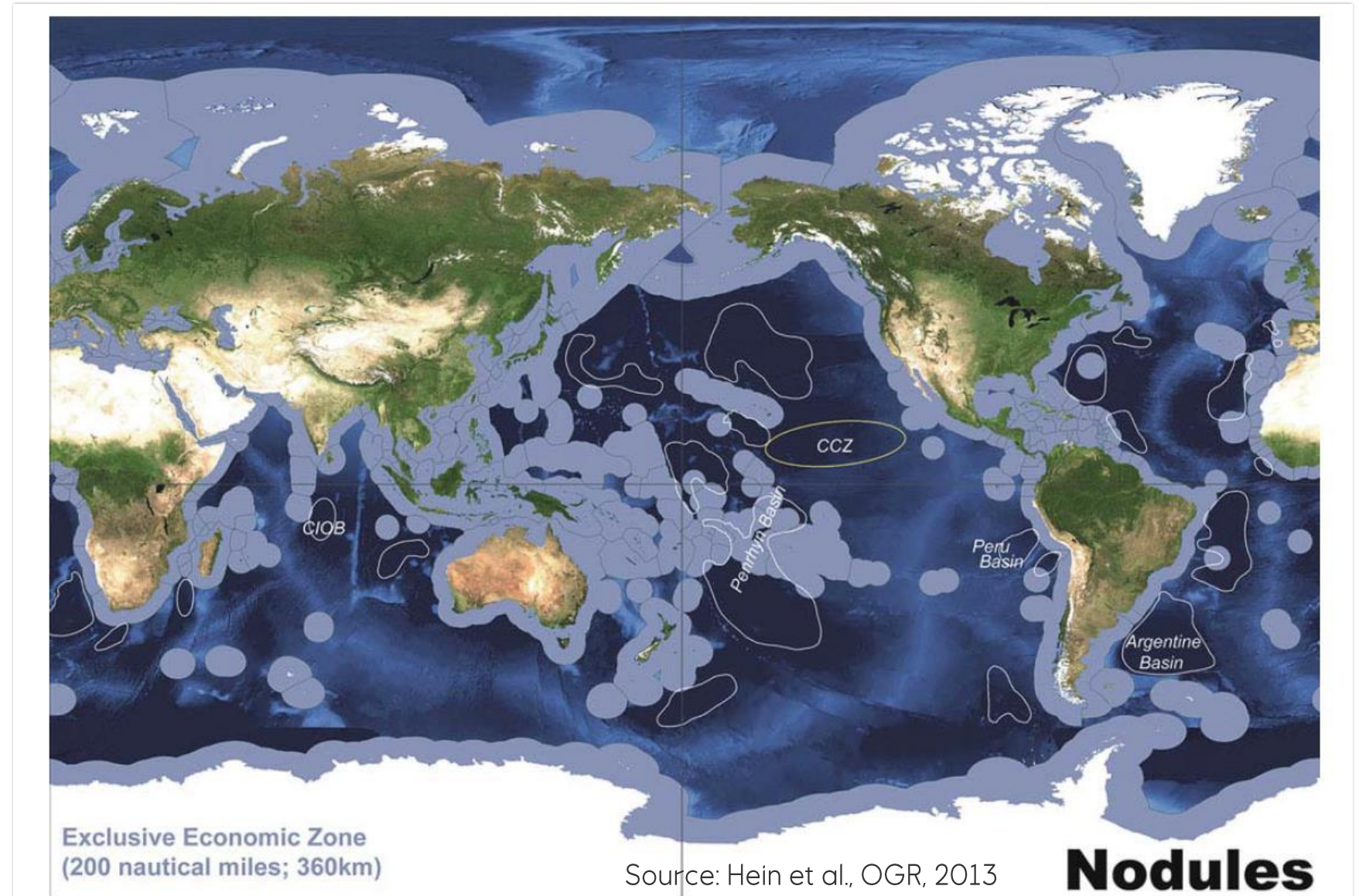
Box corer

Pictures from Wikipedia

Where are they found?

The Big Four:

- Clarion Clipperton Zone (15 kg/m²)
- Indian Ocean (5 kg/m²)
- Penrhyn Basin (5 kg/m²)
- Peru Basin (10 kg/m²)



International Seabed Authority (ISA)

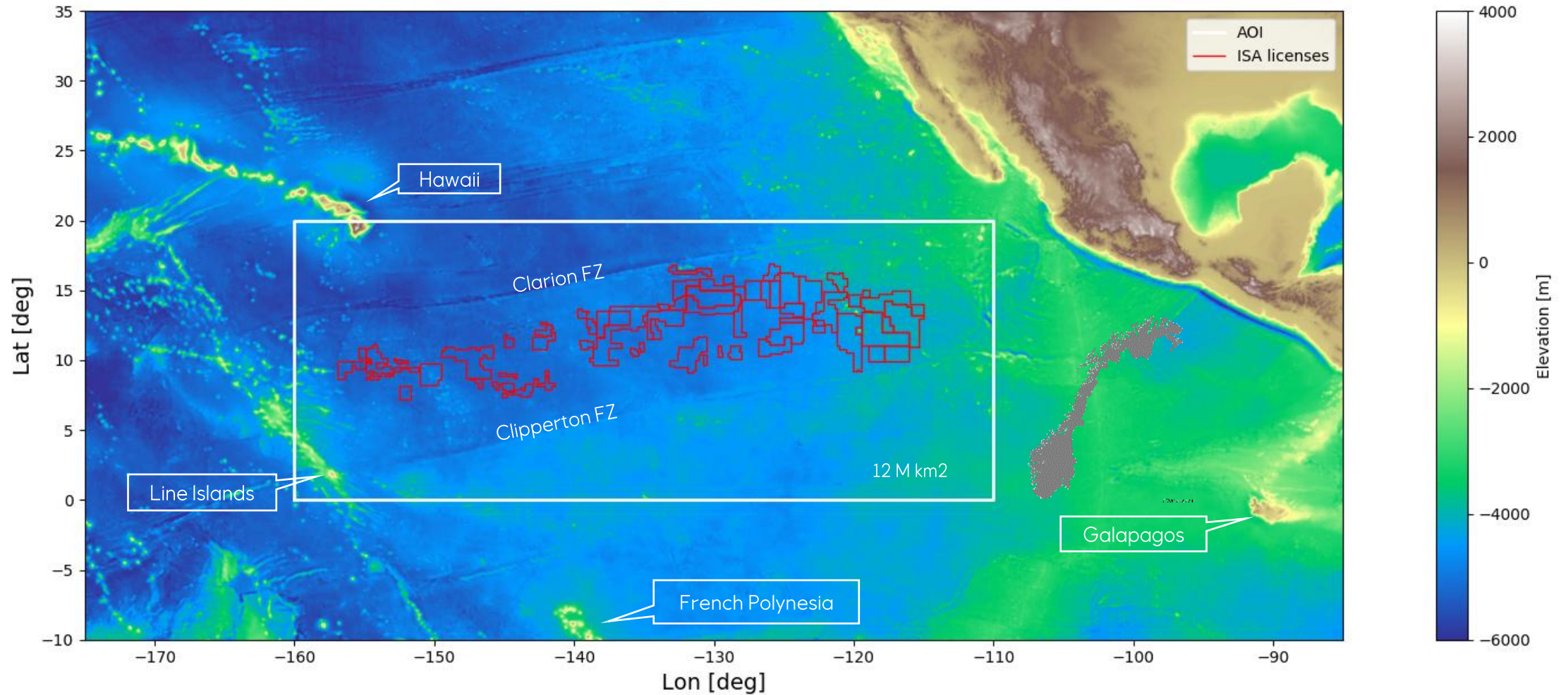
- Intergovernmental organization
- Organizing and regulating marine mining in international waters
- United Nations Convention on the Law of the Sea (UNCLOS)
 - ~170 member countries
- Awarding exploration contracts (and eventually exploitation contracts)
 - Sponsor state + operator company
 - Biology and environmental monitoring
 - Geology and resource assessment
 - Mining pilots

web site: isa.org.jm



ISA Head Quarter, Kingston, Jamaica

Clarion-Clipperton Zone



Nodule-formation model

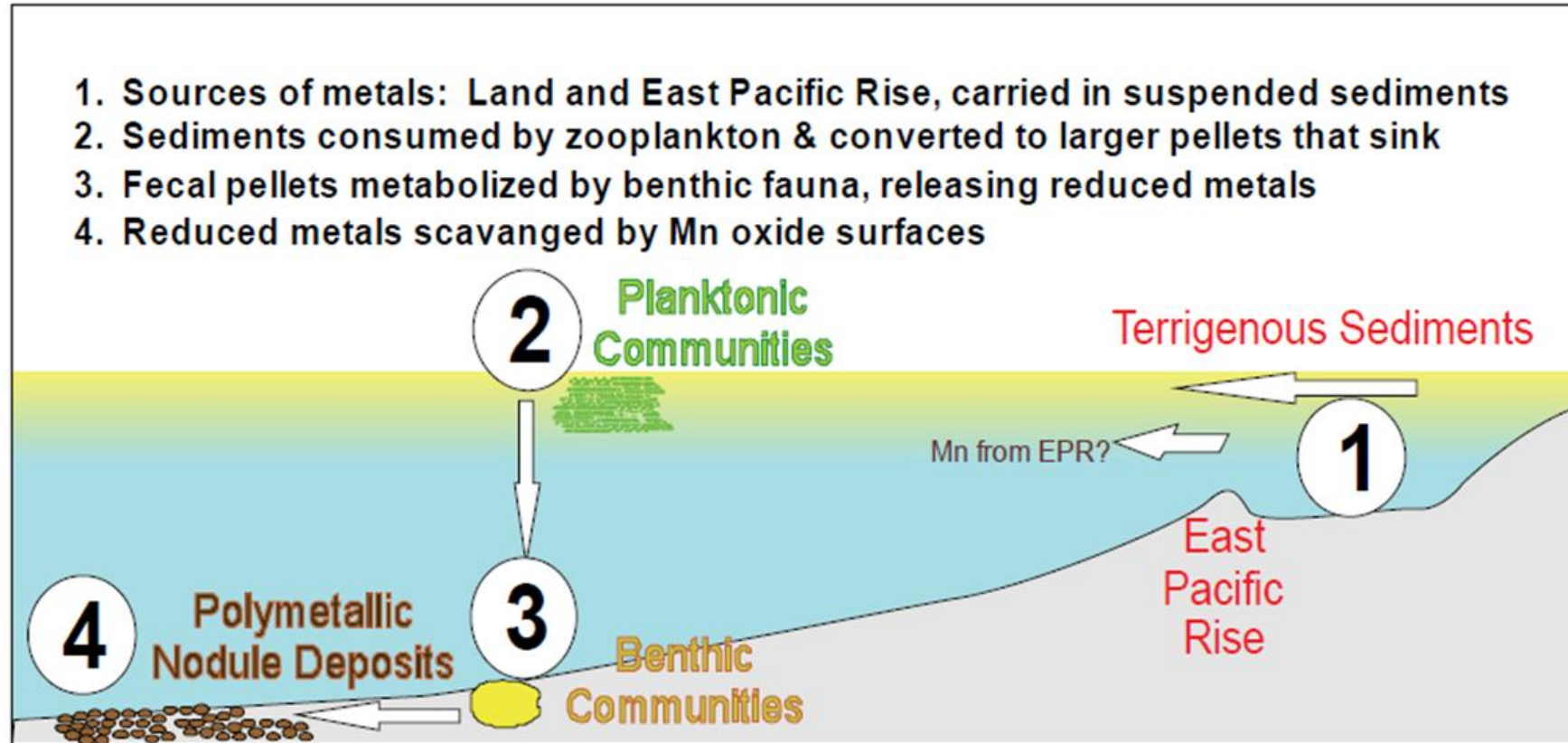


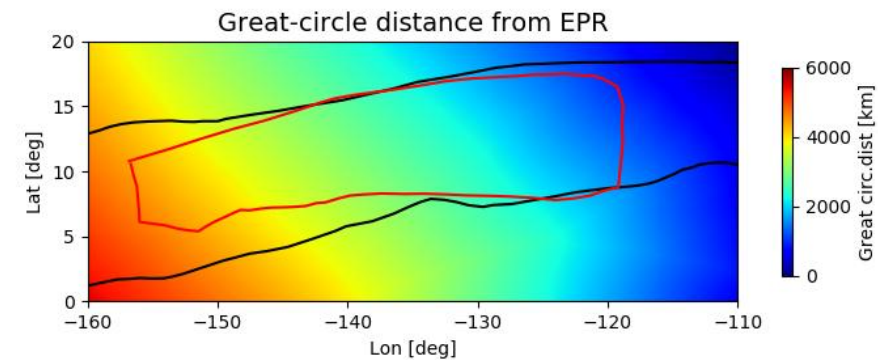
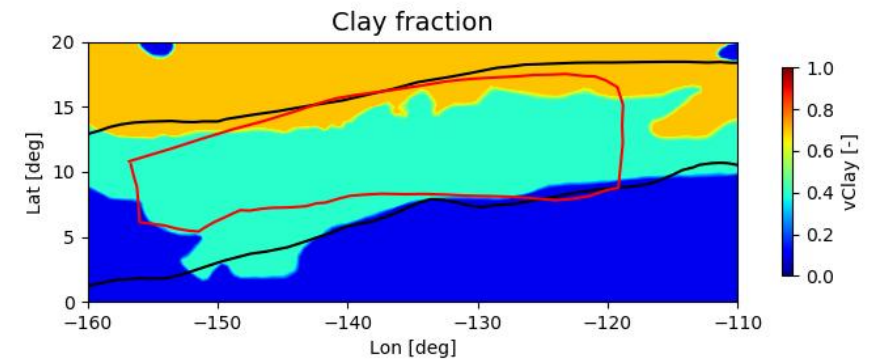
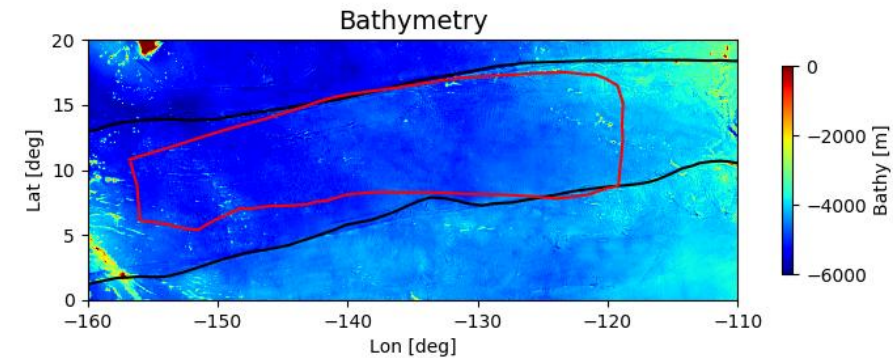
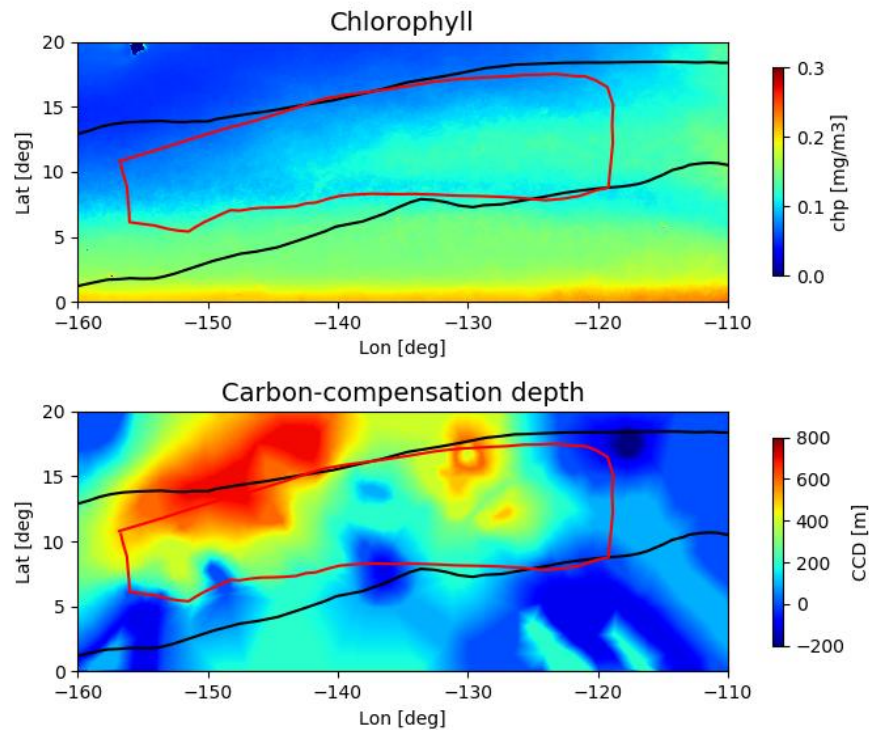
Figure from ISA report:
«A geological model of
polymetallic nodule
deposits in the CCZ»

- This model is responsible for approximately 60-70% of the polymetallic nodules in the Clarion-Clipperton Zone (Ni, Cu, Mn)
- The last 30-40% are hydrogenetic (Co, REE)

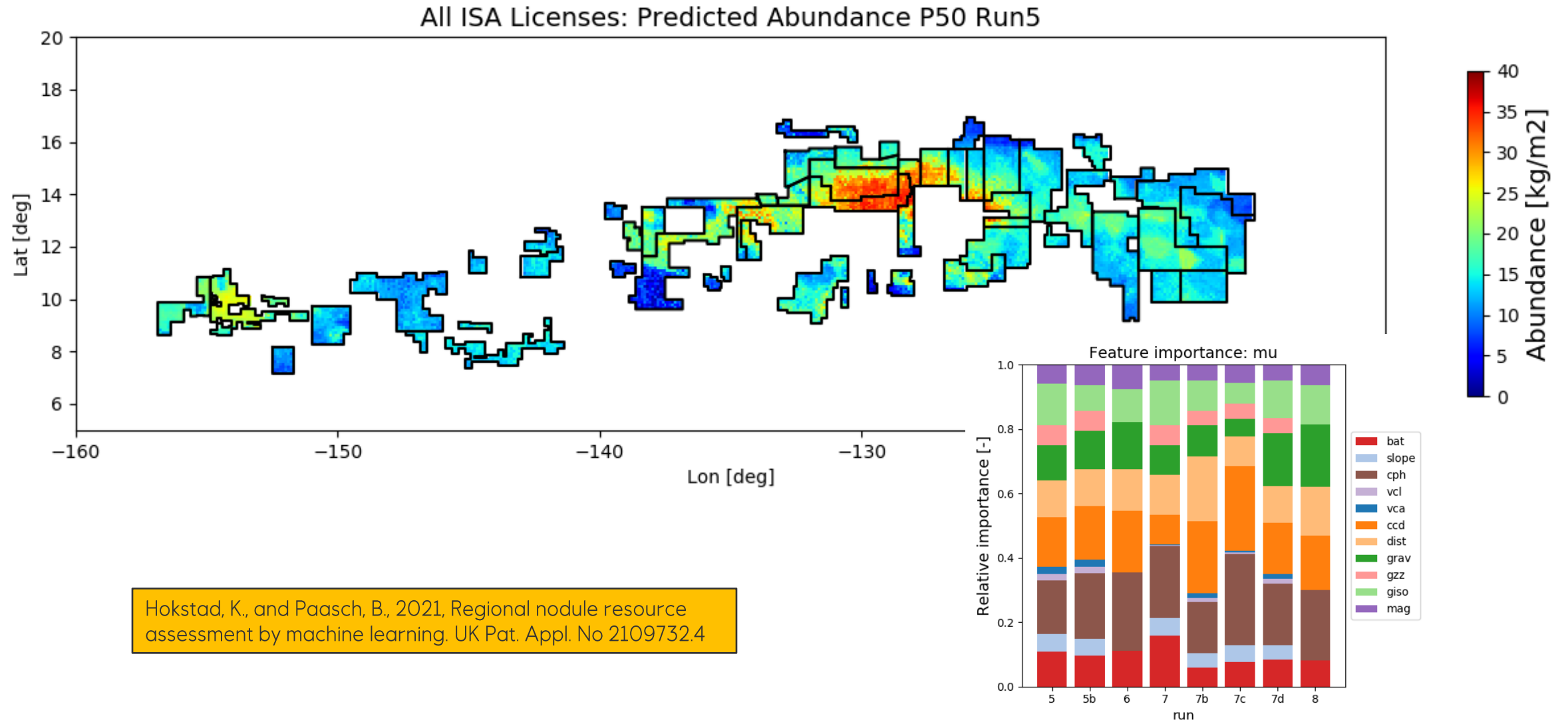
Input to machine learning

Polymetallic nodules play elements

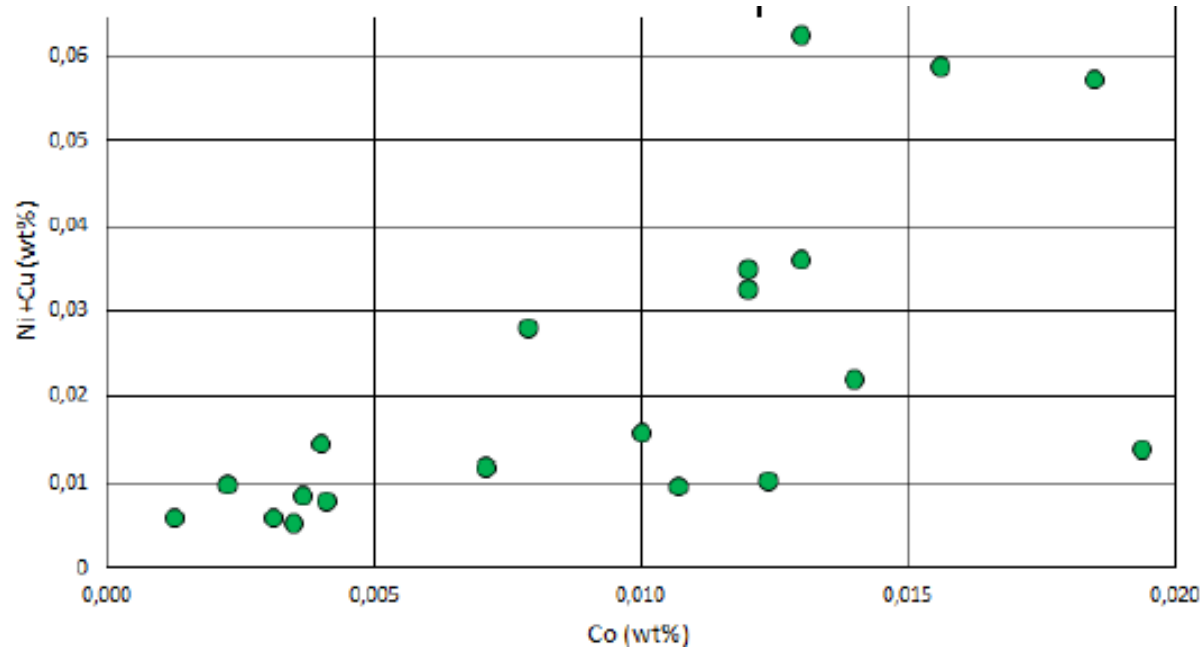
- Metal source
- Metal transport
- Favorable conditions for nodule growth



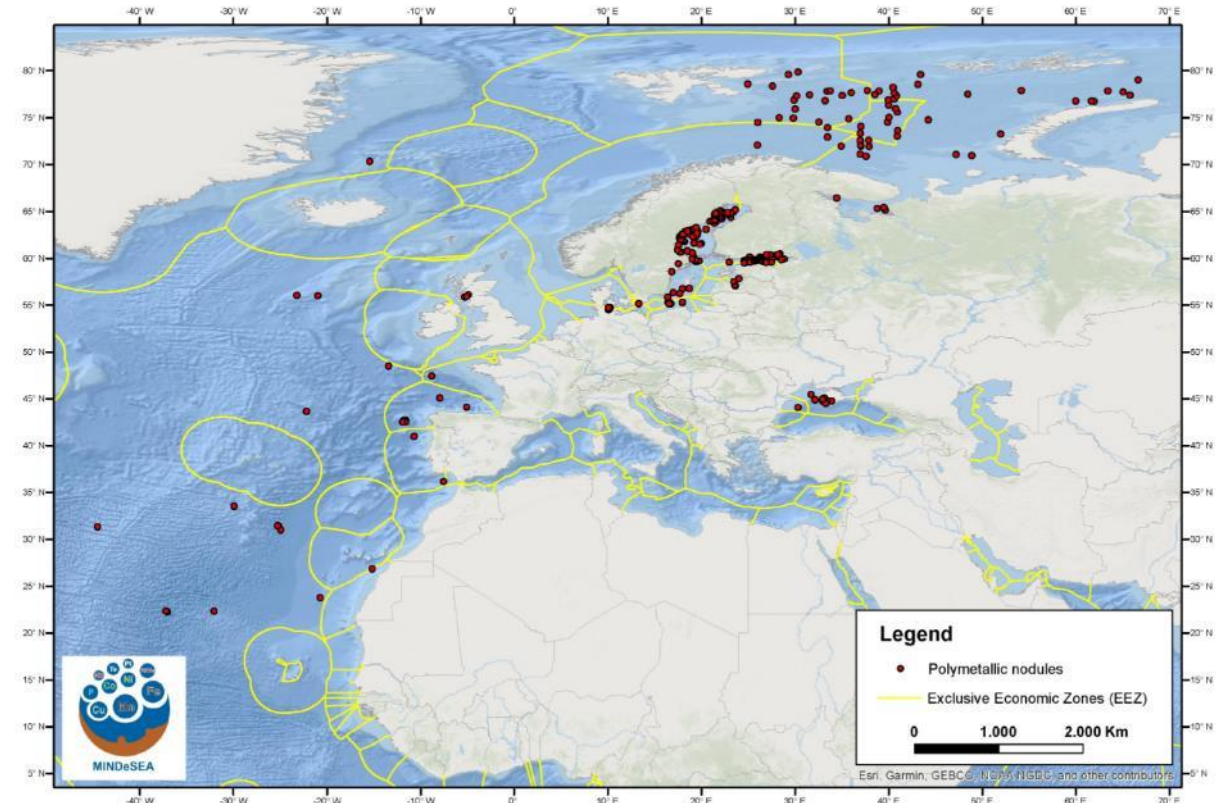
ISA CCZ licenses



Nodules in Europe

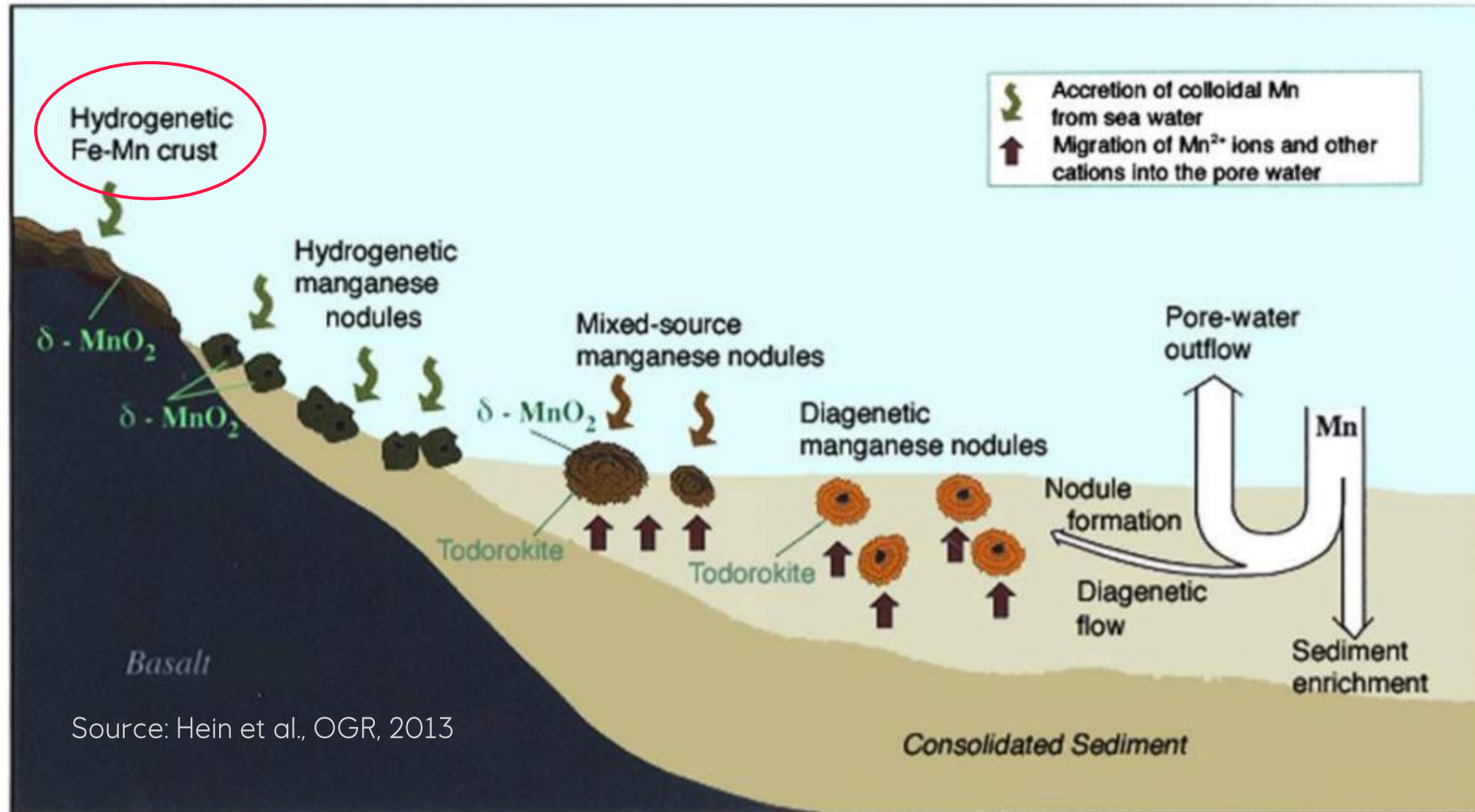


Baltic Sea: Low Cu, Ni, Co grades

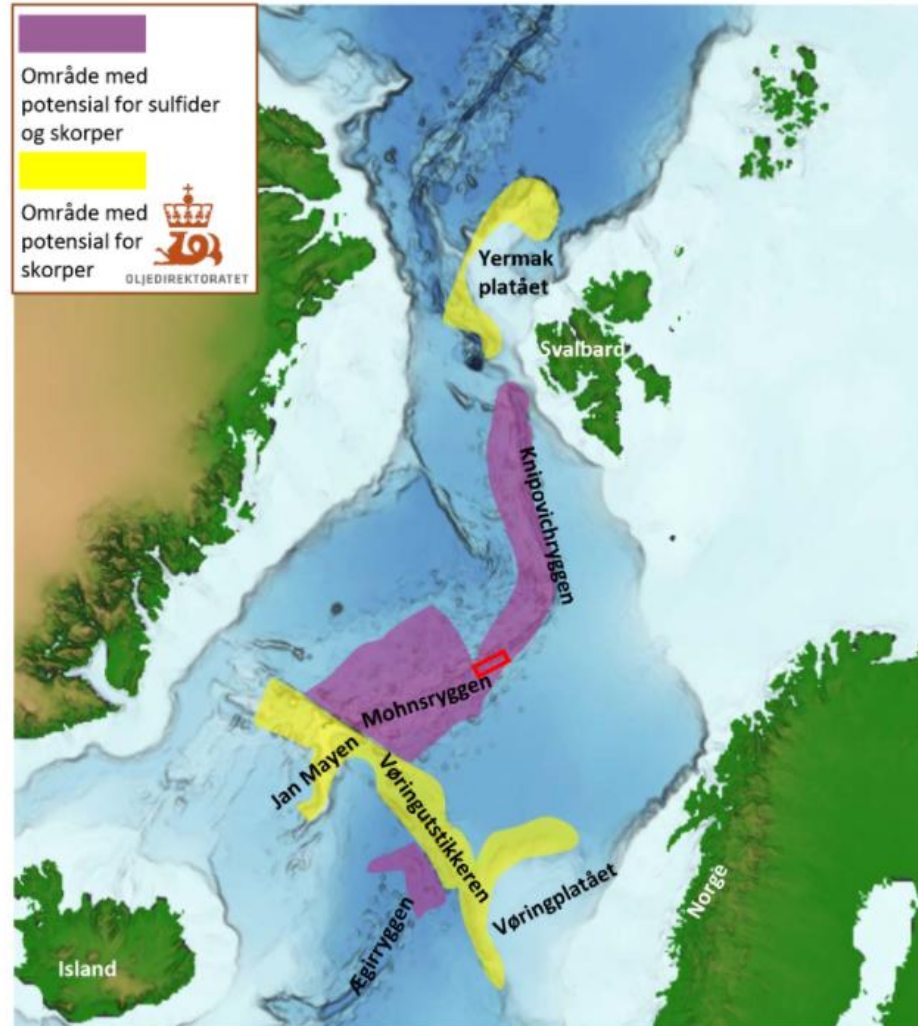


Source: MINDeSEA (WP 6) EU Project

Ferro-manganese crusts



Ferro-manganese crusts on NEEZ

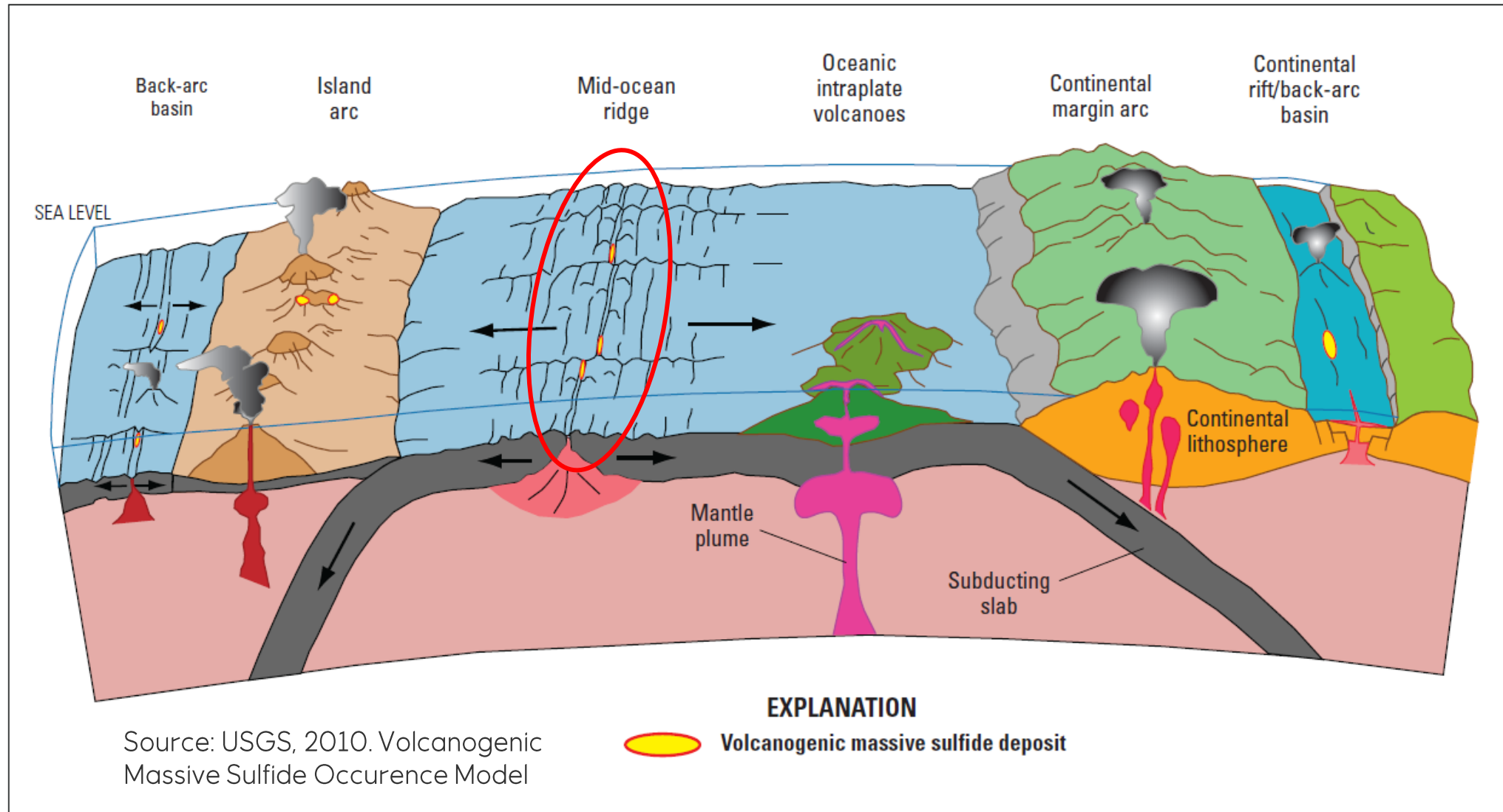


- Metals precipitated from cold sea water
- Similar to nodules (hydrogenetic)
- Growing on seamounts and ridges
- Slow growth rate $< 1 \text{ cm/Ma}$



Picture from NPD (www.oljedirektoratet.no)

Seabed massive sulfides (SMS)

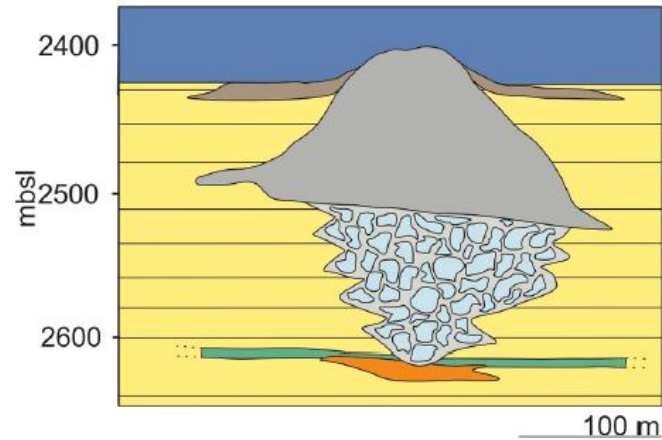


Løkken, Trøndelag

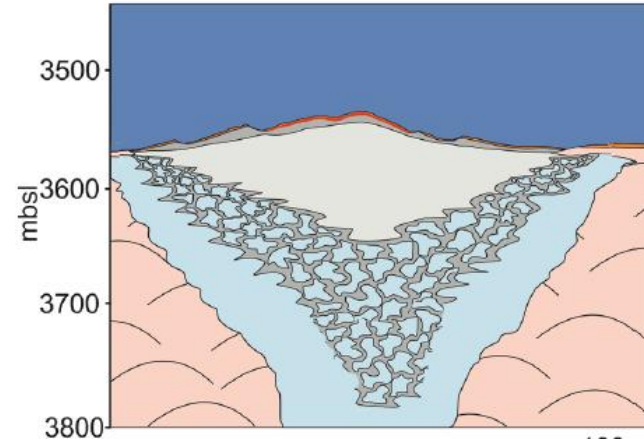
Løkken, Trøndelag

Three interesting examples for geophysicists

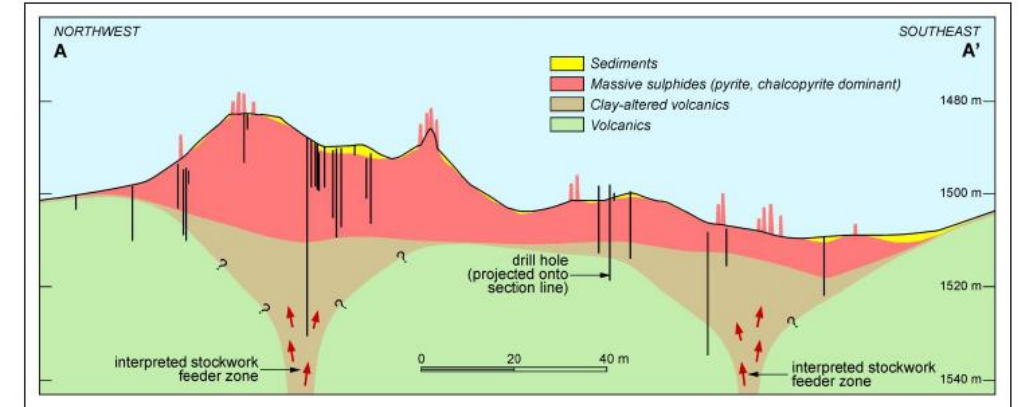
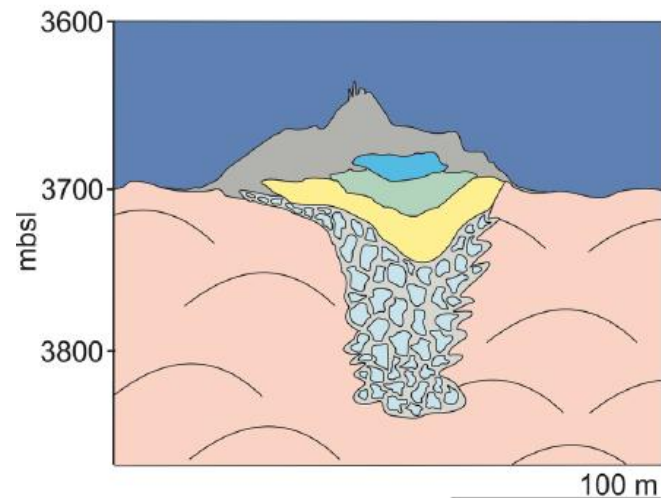
C. Bent Hill, Middle Valley, Juan de Fuca Spreading Centre



A. Inactive Southern Mound, TAG Hydrothermal Field, MAR

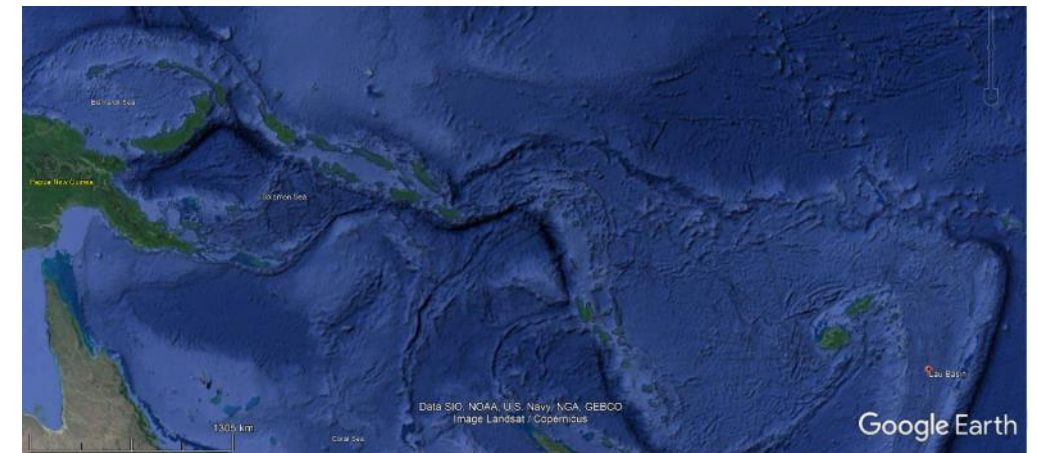


B. TAG Active Mound, MAR



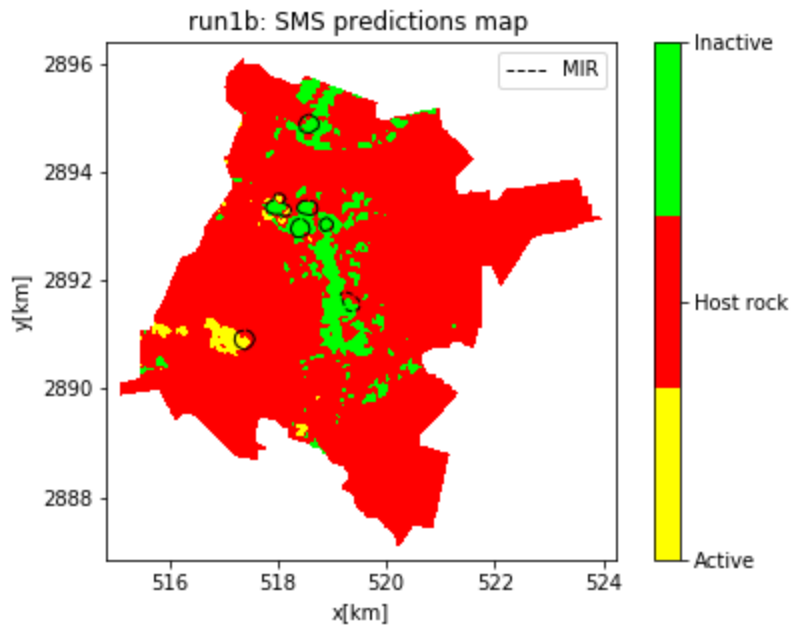
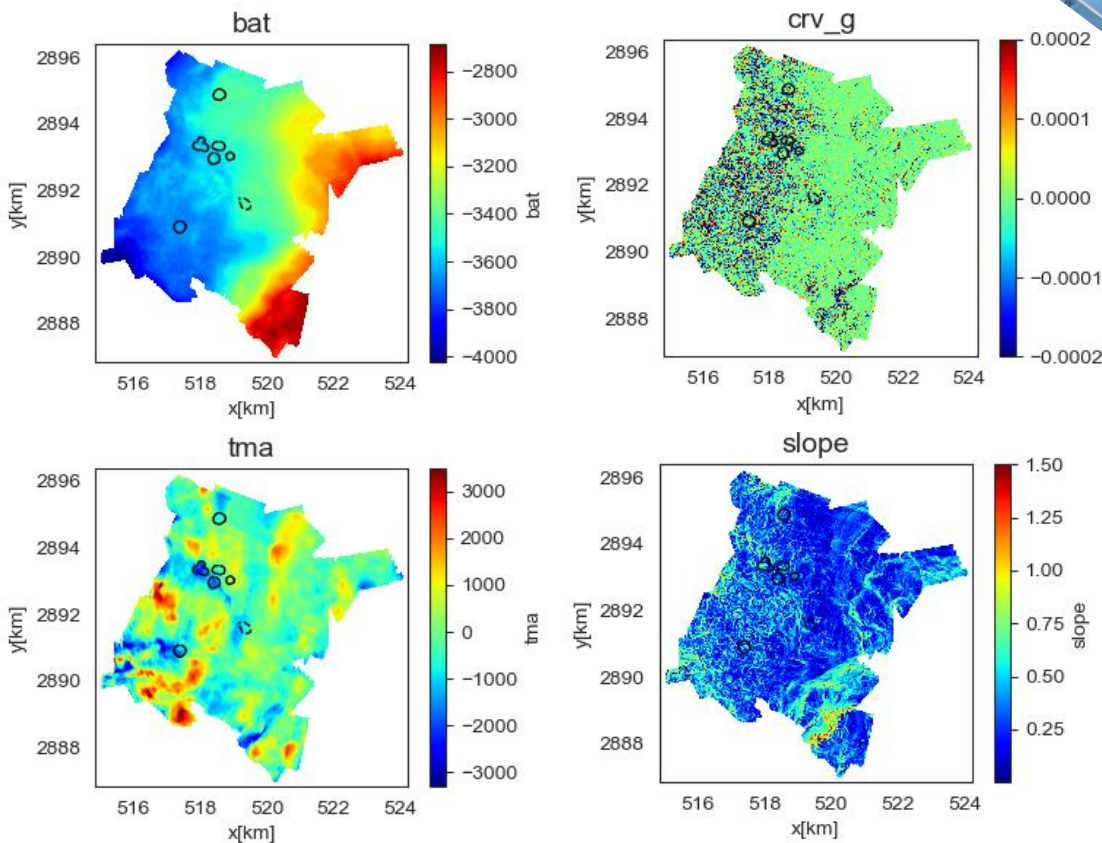
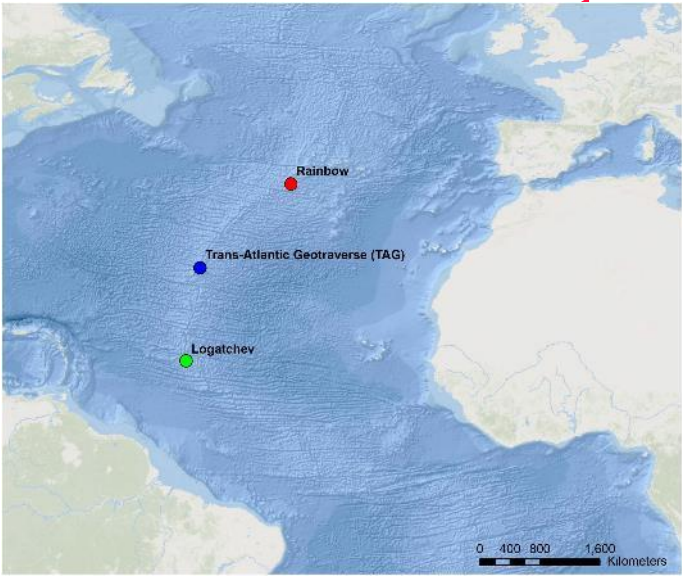
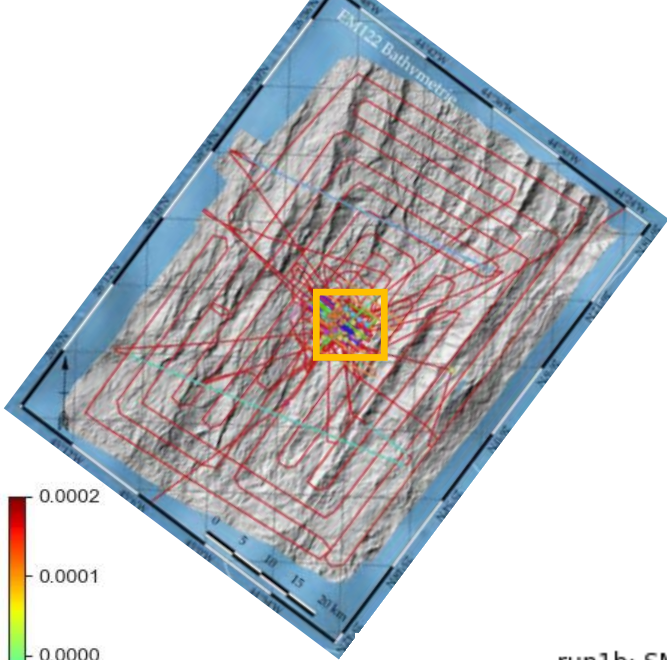
Source: Nautilus Minerals, Solwara-1 PEA

Source: Murton et al., 2019.
Geological fate of seafloor massive sulfides at the TAG hydrothermal field. OGR



Transatlantic geotraverse (TAG) ML using MBES and magnetics

Figure from Petersen et al.,
M127 Cruise Report, 2007



Hokstad, K., and Paasch, B., 2021, Regional nodule resource
assessment by machine learning. UK Pat. Appl. No 2109732.4

Public data from M127 cruise (pangea.de)

Reflection and refraction imaging with site survey vessel

Source 1:

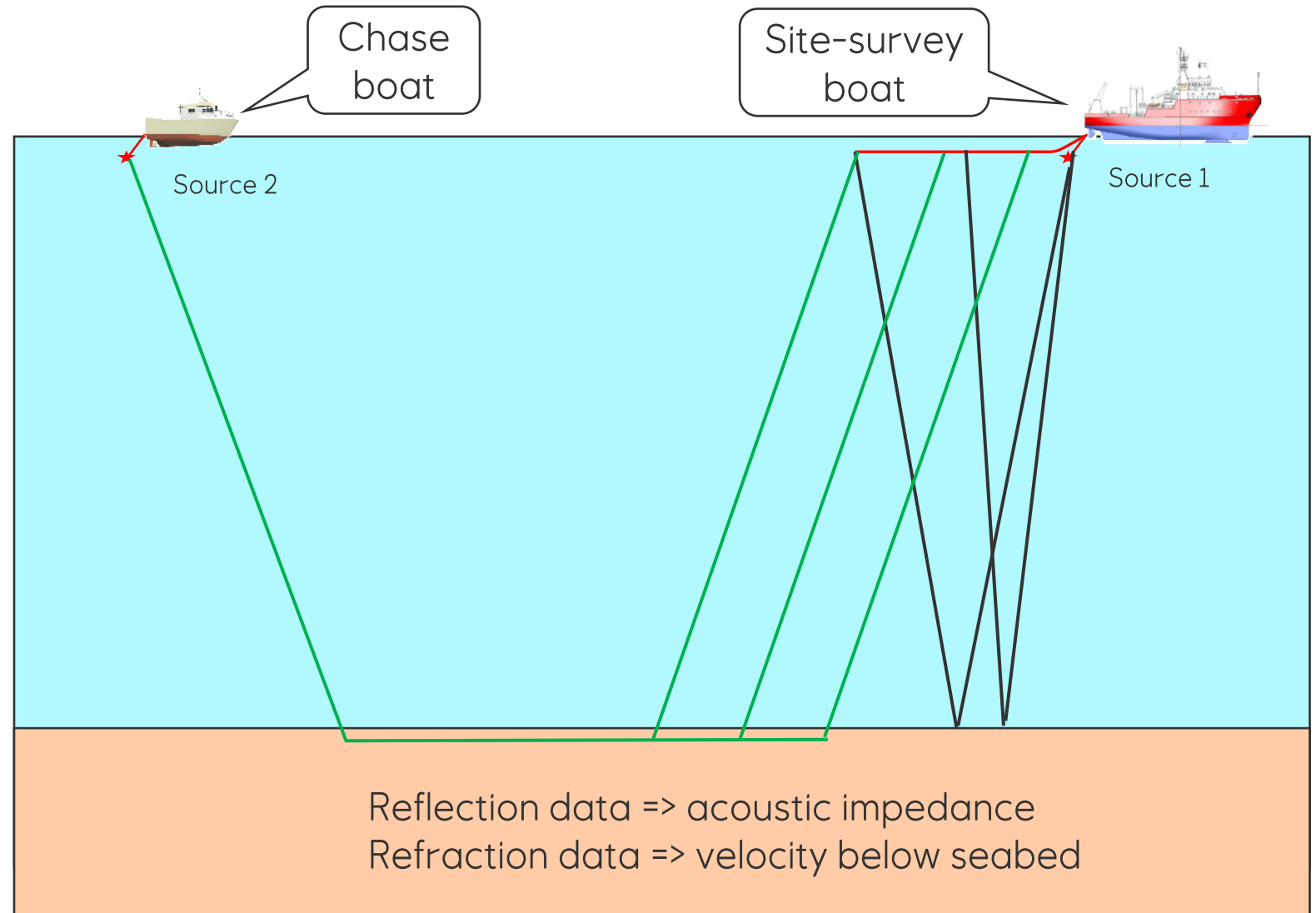
- Short-spread receiver tow
- Acoustic impedance

Source2:

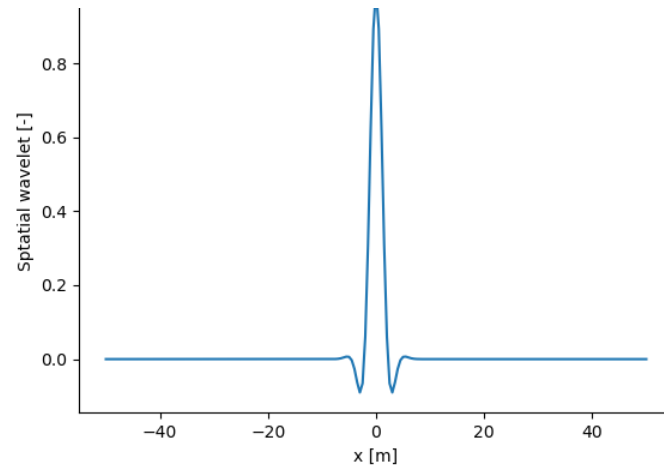
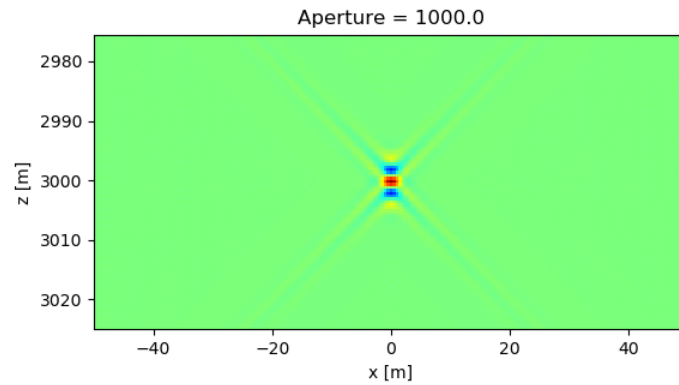
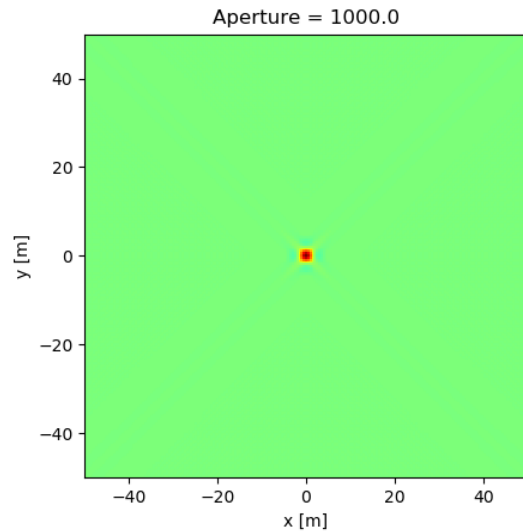
- Needed for refraction data
- Velocity below seabed

Site survey-vessels also record:

- Multi-beam echosounder
- Sub-bottom profiler (?)

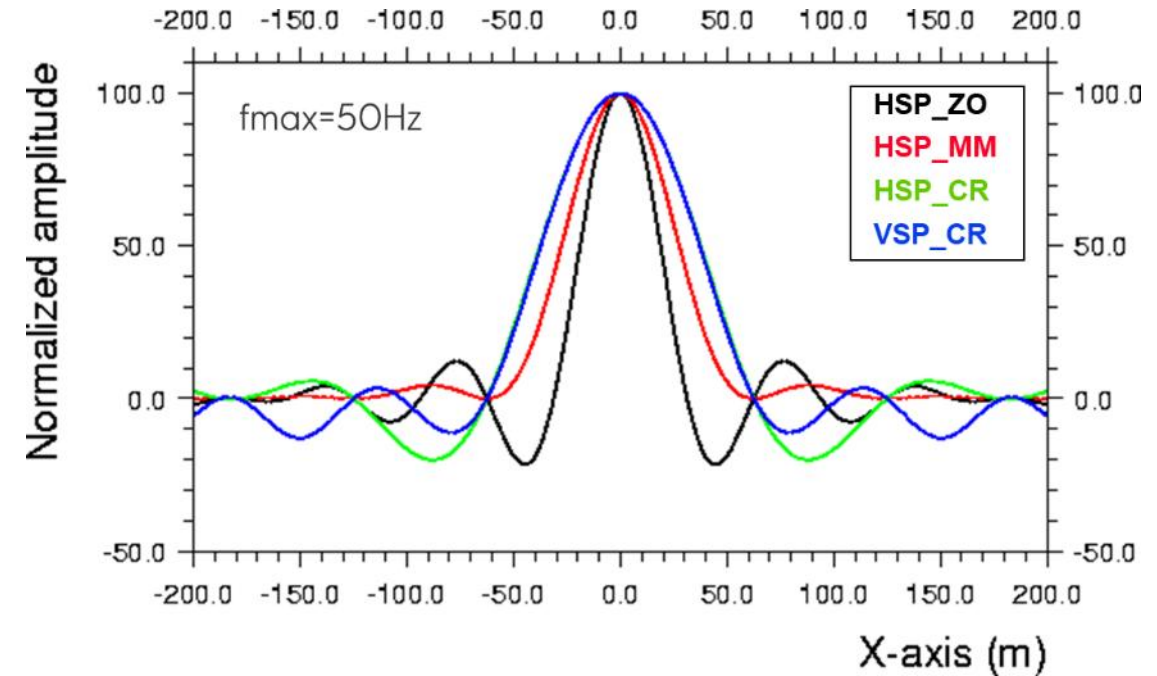


The seismic image of a diffractor – spatial resolution



Analytic migration in the Fraunhofer approximation

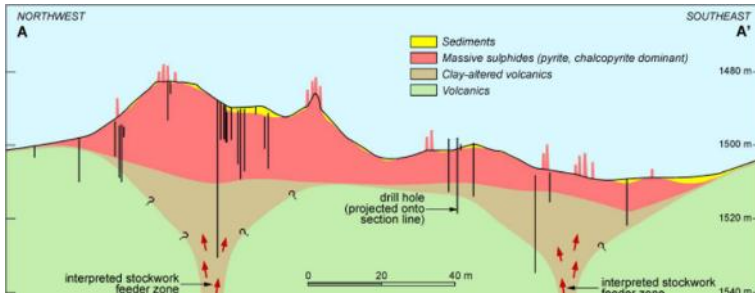
- $f_{max} = 250$ Hz
- depth = 2500m



- The best spatial resolution is obtained from zero-offset data (Vermeer, 1999)

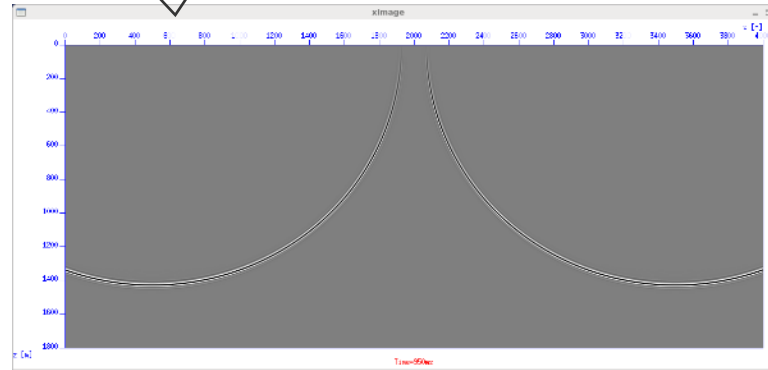
Hokstad, Sollie and Petersen, 2001, Horizontal Resolution of 3D Walkaway VSP Data, SEG

Elastic finite-difference feasibility modeling

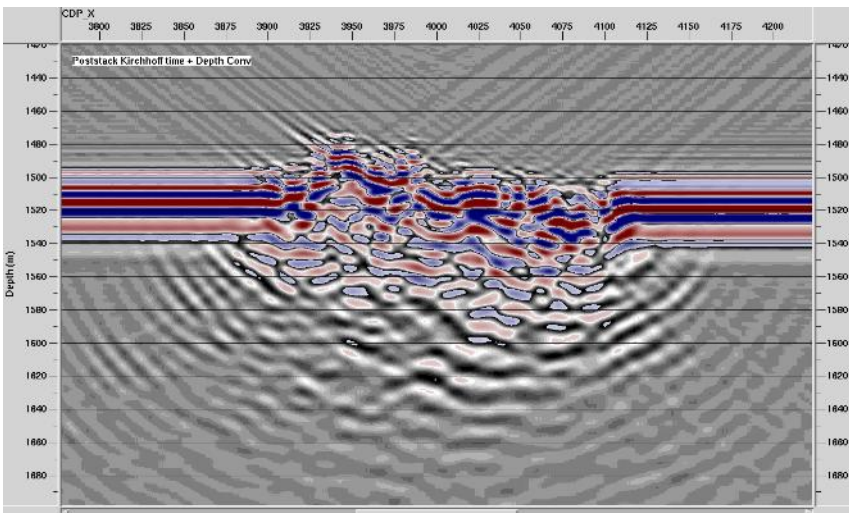
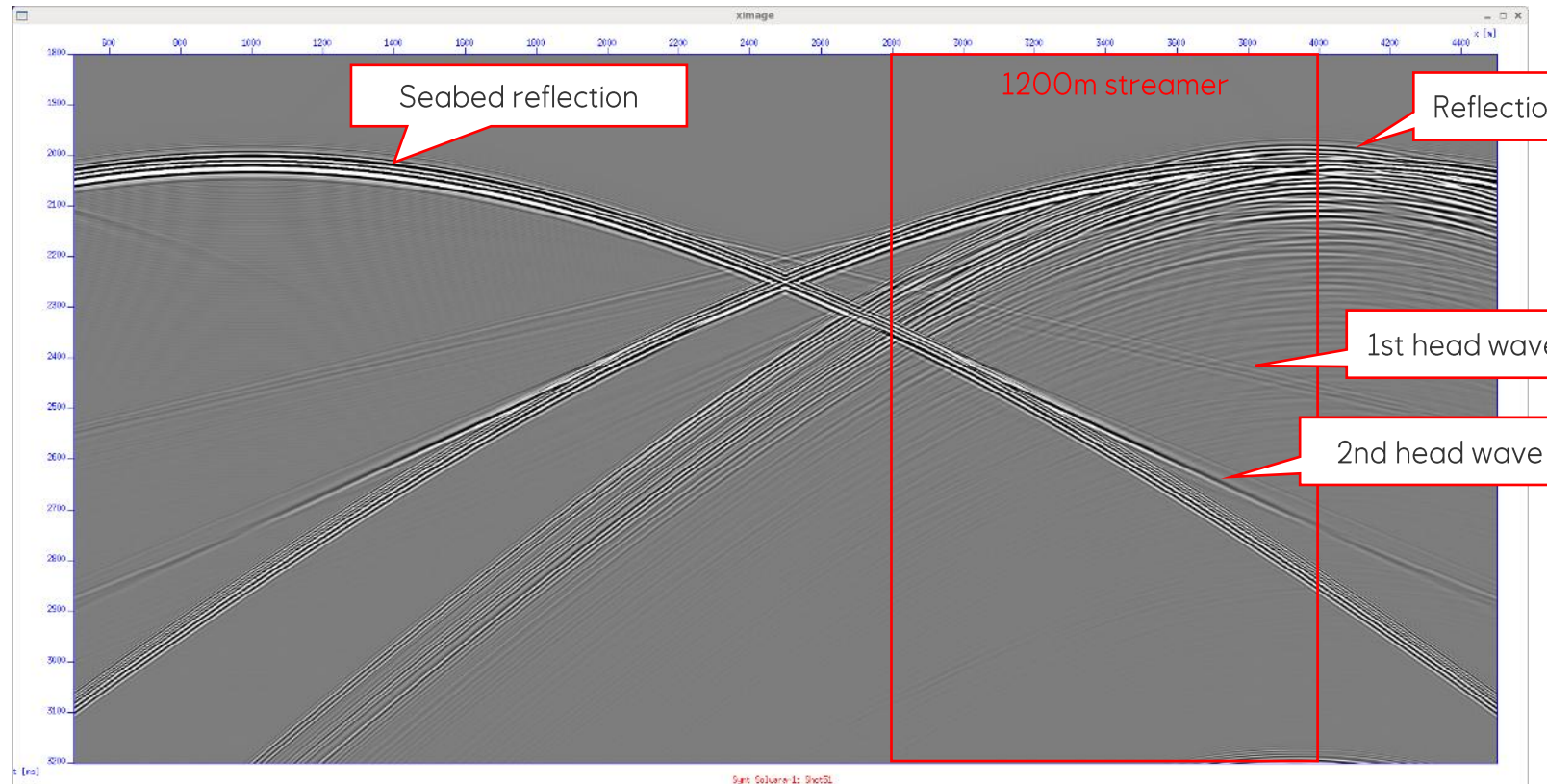
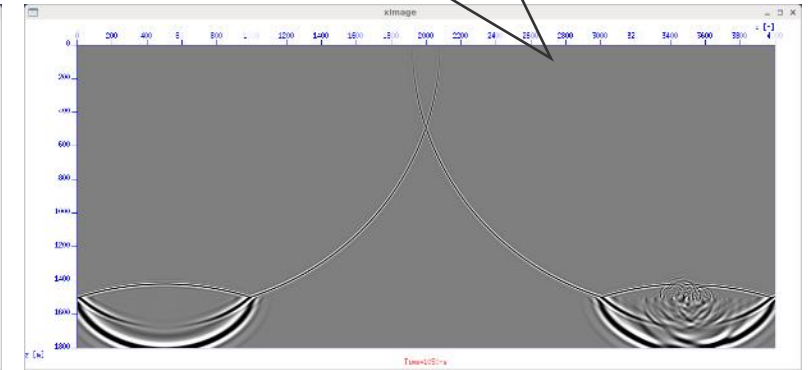


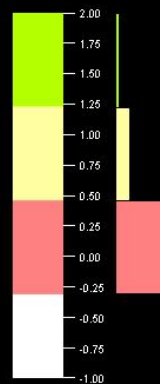
- Geometry from Solwara-1, Bismarck Sea
- Geophysical parameters from TAG
- Max frequency: 250 Hz

Snapshot: incident wavefield



Snapshot: Wavefield hitting the target

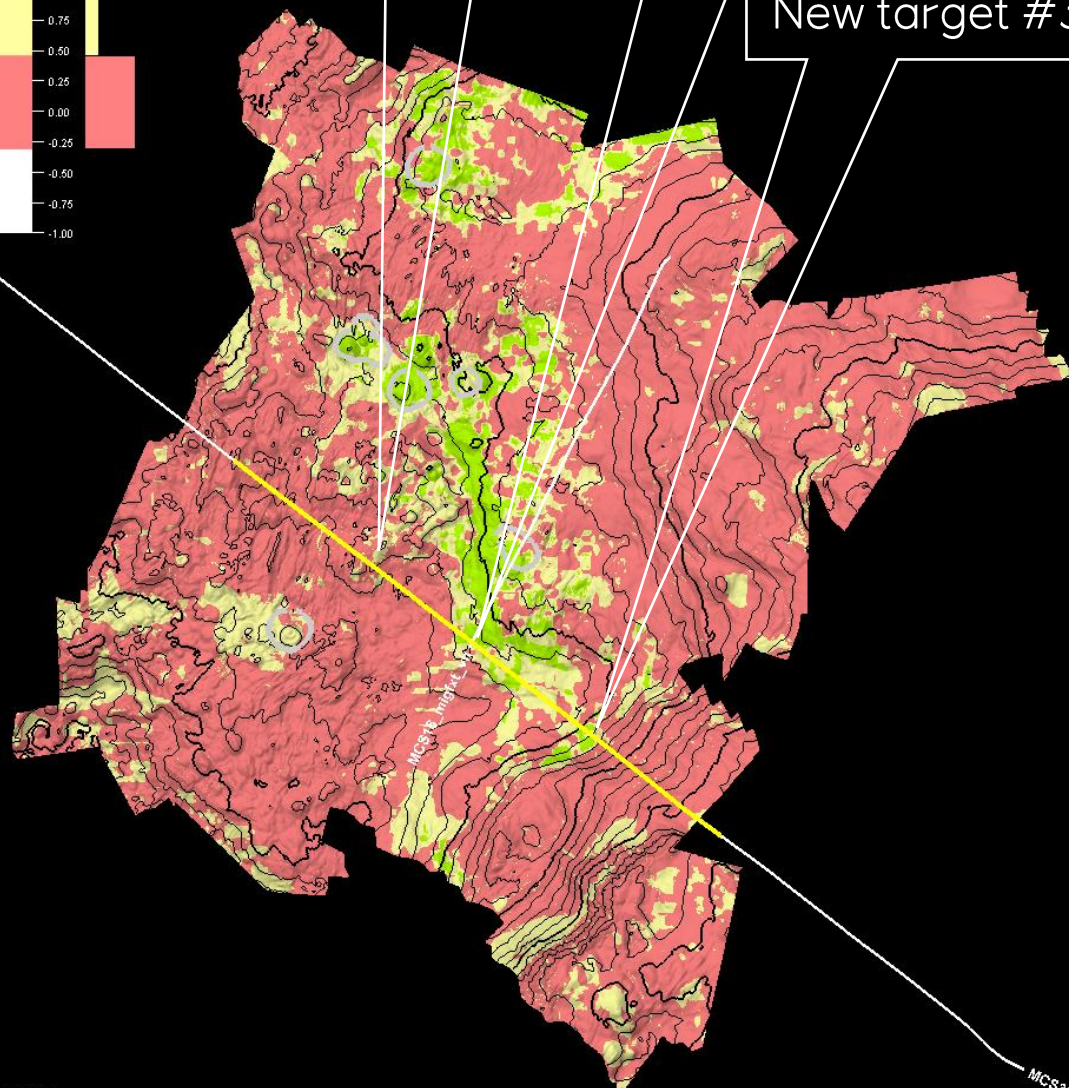




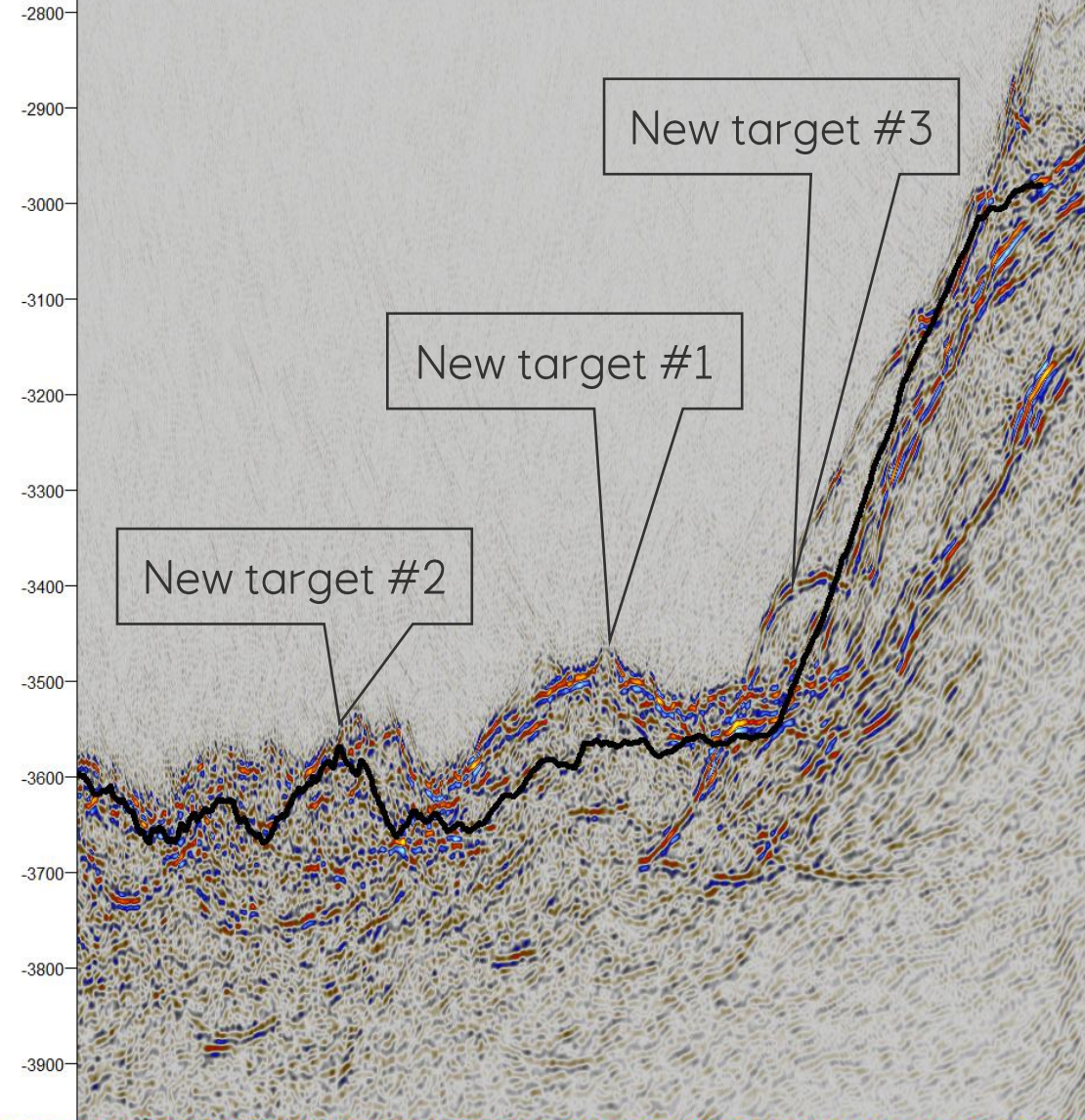
New target #2

New target #1

New target #3



2000m



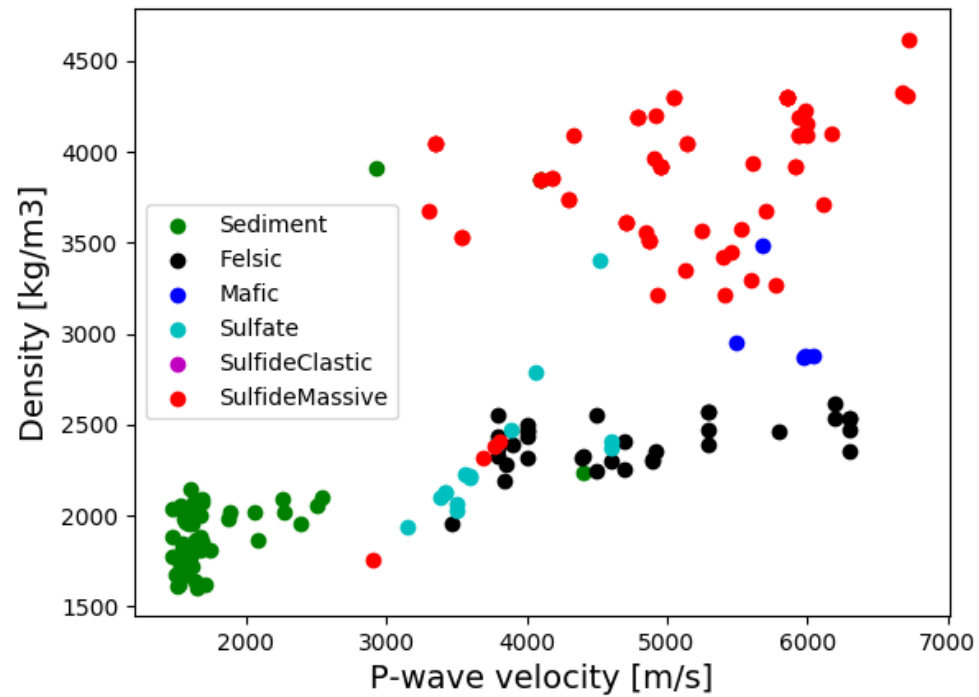
New target #1

New target #3

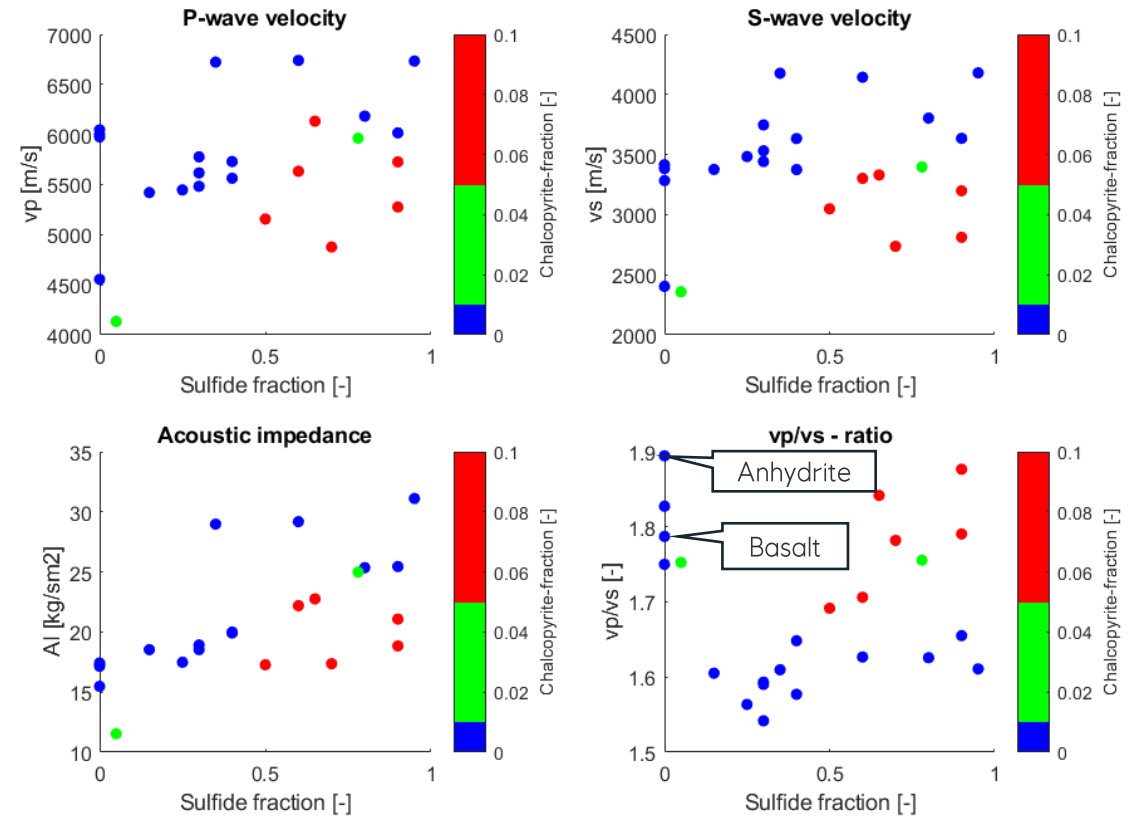


Elastic properties

TAG, BH, Manus: P-wave velocity vs density



Data from ODP Legs 158, 139/169, 193



TAG ODP Leg 158



Contents lists available at ScienceDirect

Journal of Volcanology and Geothermal Research

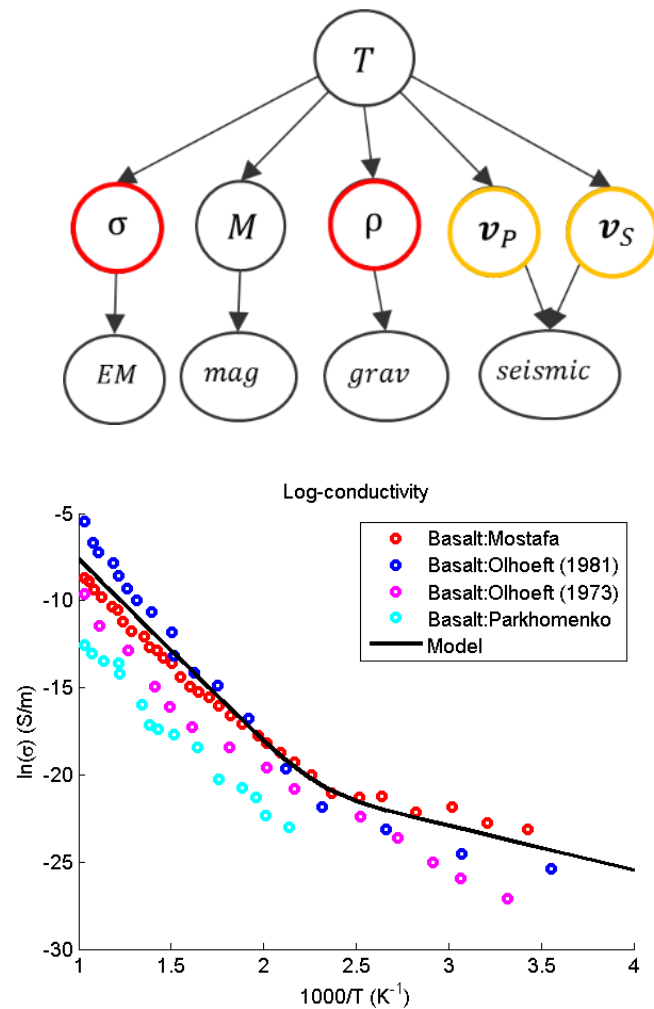
journal homepage: www.elsevier.com/locate/jvolgeores



The Iceland Deep Drilling Project at Reykjanes: Drilling into the root zone of a black smoker analog

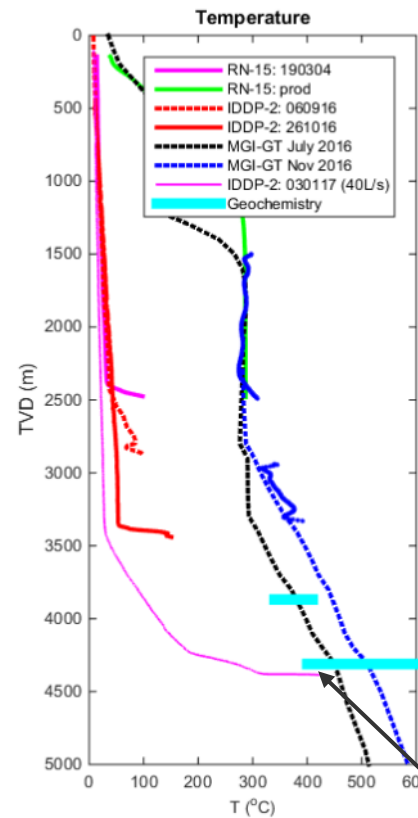
Guðmundur Ó. Friðleifsson ^{a,*}, Wilfred A. Elders ^b, Robert A. Zierenberg ^c, Andrew P.G. Fowler ^d, Tobias B. Weisenberger ^e, Kiflom G. Mesfin ^a, Ómar Sigurðsson ^a, Steinþór Níelsson ^e, Gunnlaugur Einarsson ^e, Finnbogi Óskarsson ^e, Egill Á. Guðnason ^e, Helga Tulinius ^e, Ketil Hokstad ^f, Gibert Benoit ^g, Frank Nono ^{g,h}, Didier Loggia ^g, Fleurice Parat ^g, Sarah B. Cichy ^{g,i}, David Escobedo ^g, David Mainprice ^g

Geothermal exploration: Reykjanes, Iceland



TD (4659 mMD):

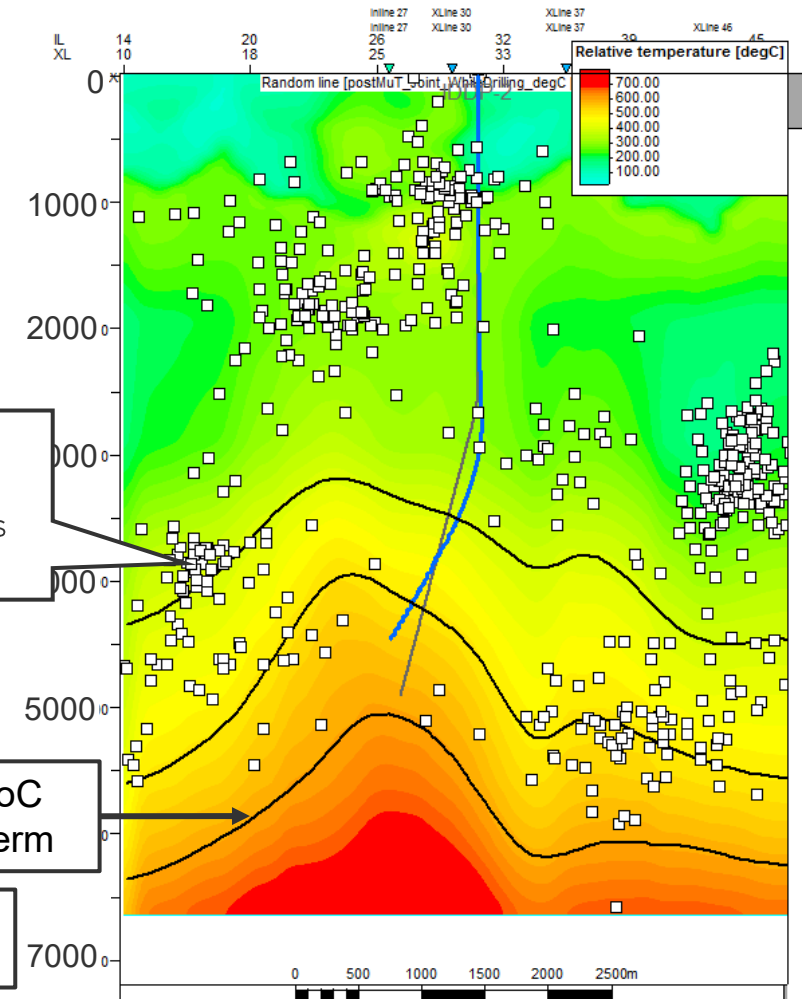
- Inversion: 535 ± 50 °C
- Horner plot: ~ 530 °C



□ Earthquake locations
No earthquakes in ductile zone

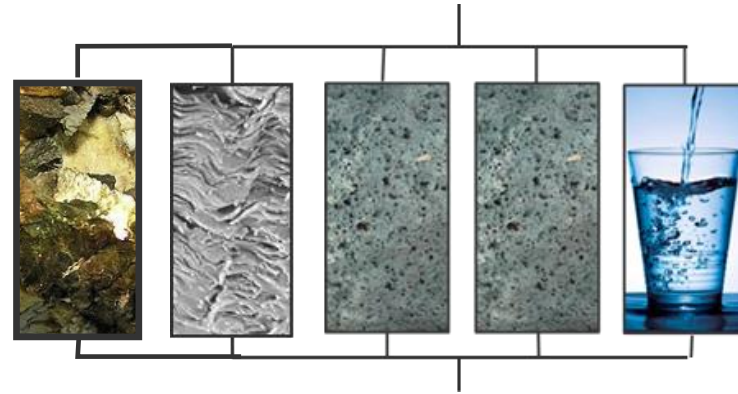
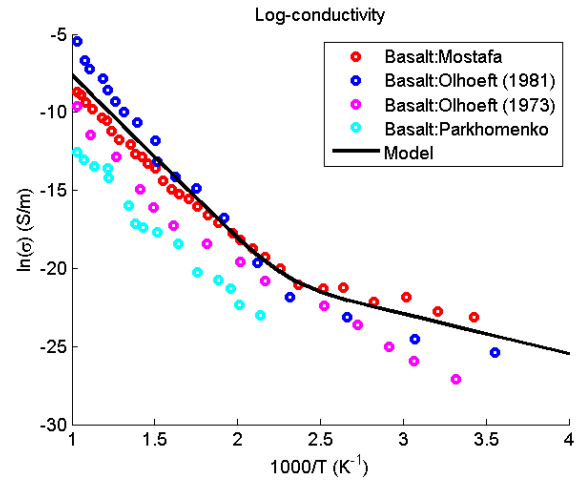
600 oC isotherm

3. January 2017:
Measured 426 °C

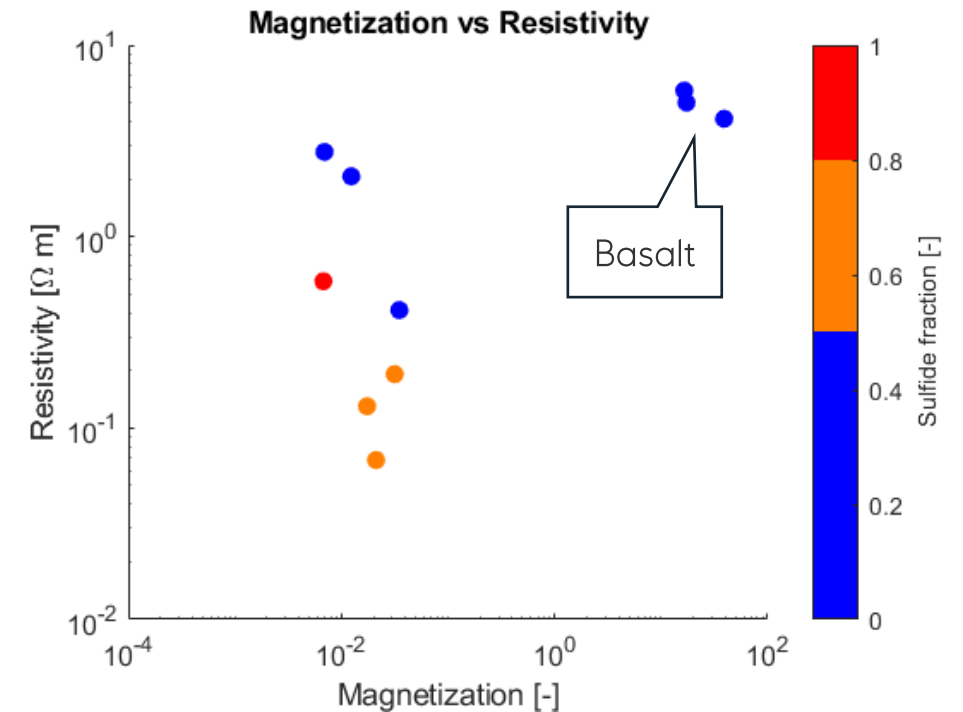
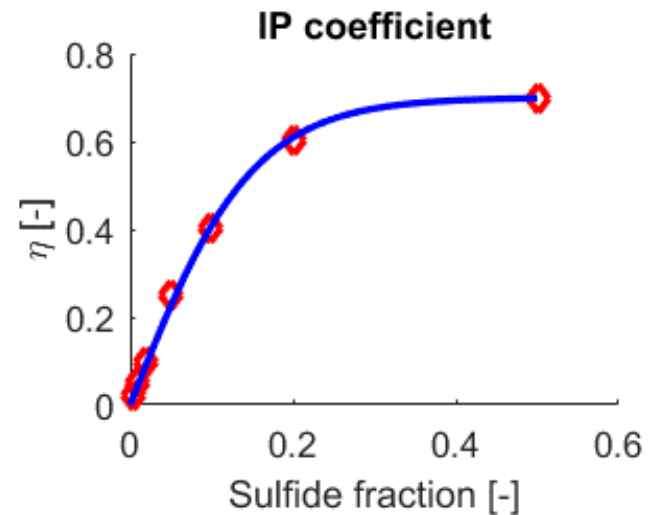


Hokstad, K. and K. Tānavsuu-Milkeviciene, 2017. Temperature Prediction by Multigeophysical Inversion: Application to the IDDP-2 Well at Reykjanes, Iceland, Geothermal Resource Council, Transactions, 42, 1141-1152.

Electromagnetic properties

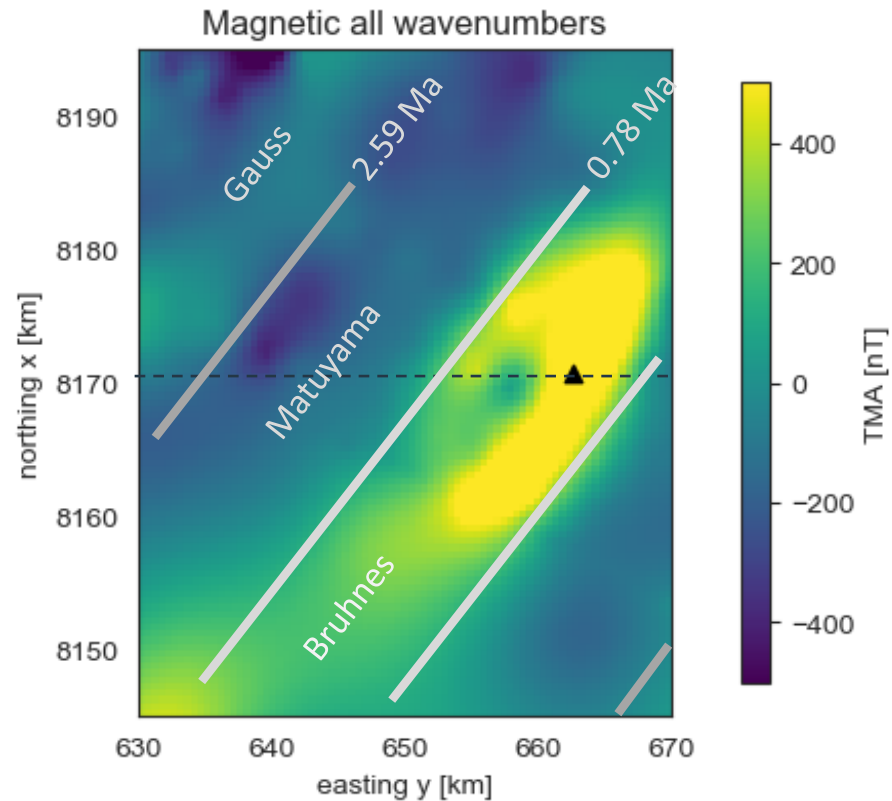


- Parallel coupling with
 - Basalt (2 mechanisms)
 - Clay (alteration minerals)
 - Pore-water
 - Metal sulfides

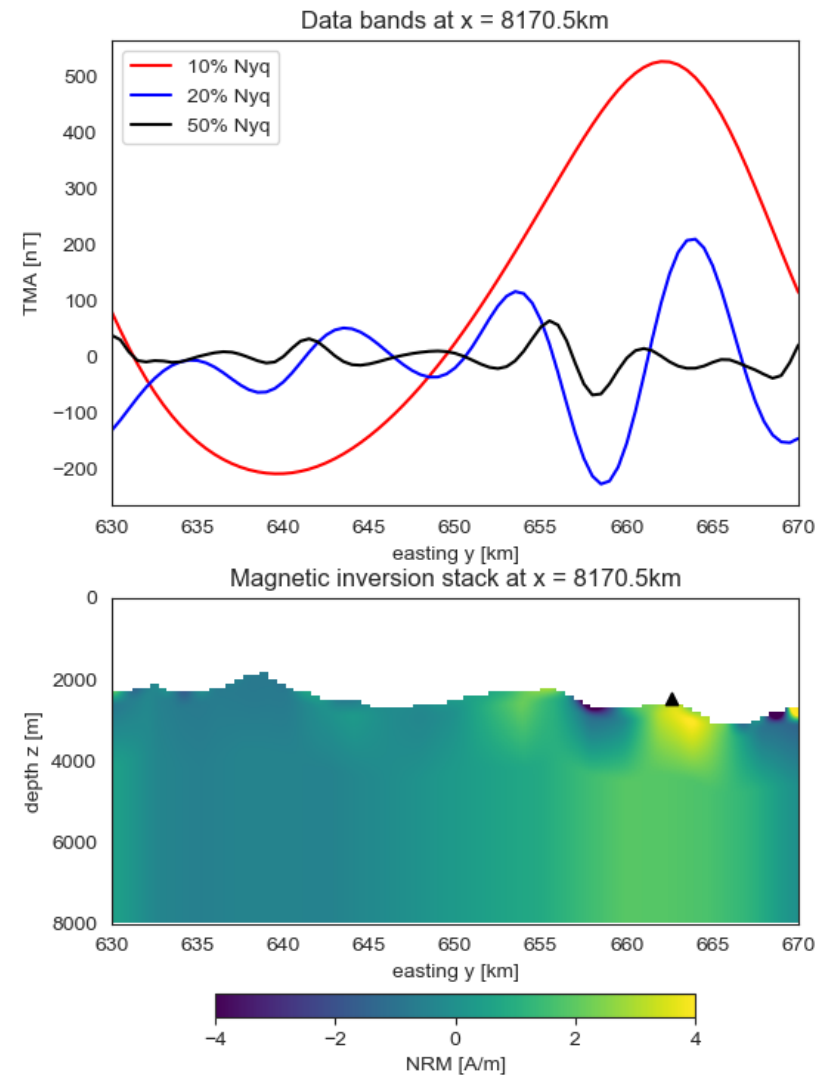


ODP Leg 158 (TAG)

Mohn's Ridge aeromagnetic data



Spreading rate: 15 mm/a



SMS bodies not detectable with aeromag (?)

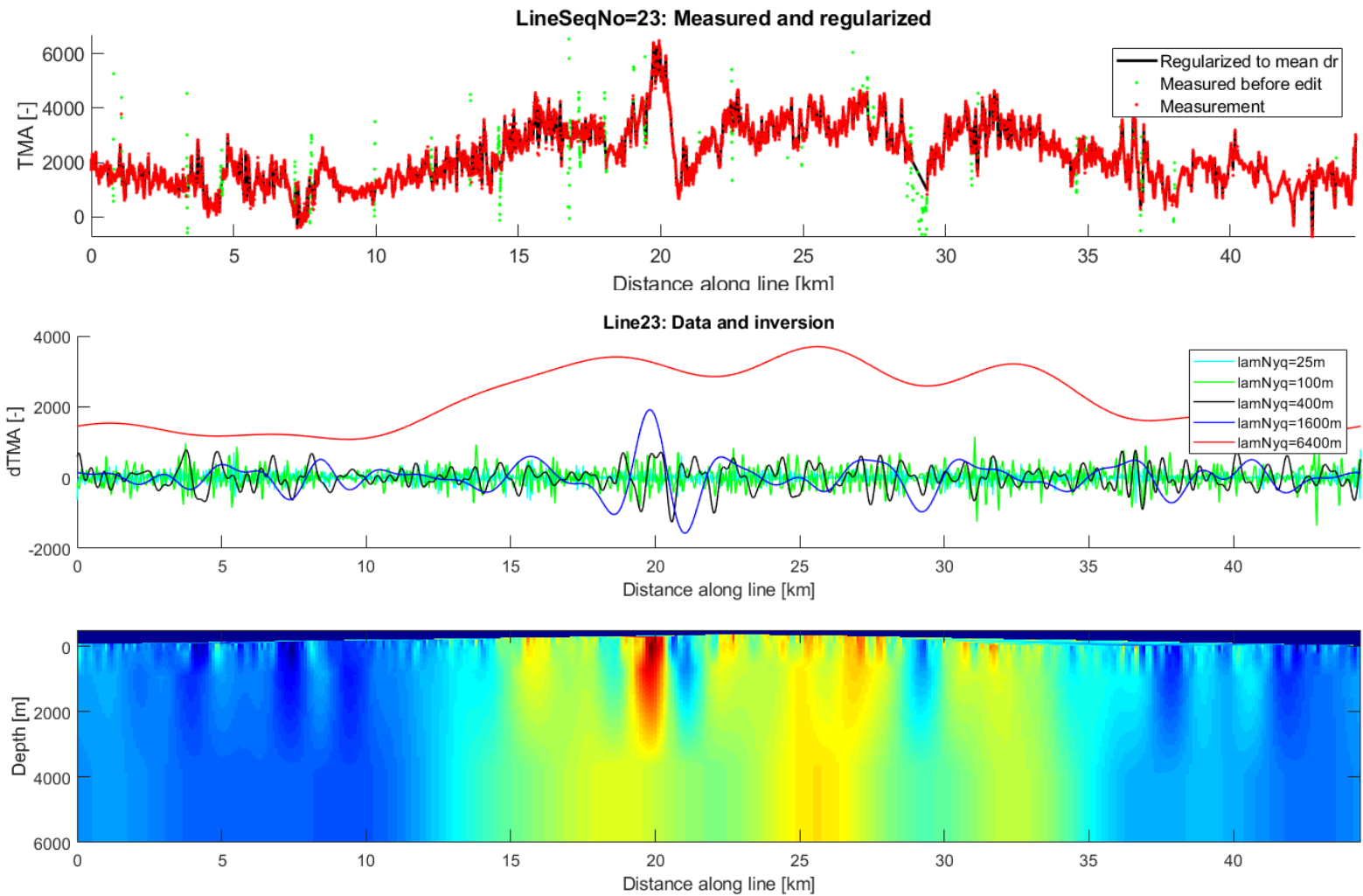
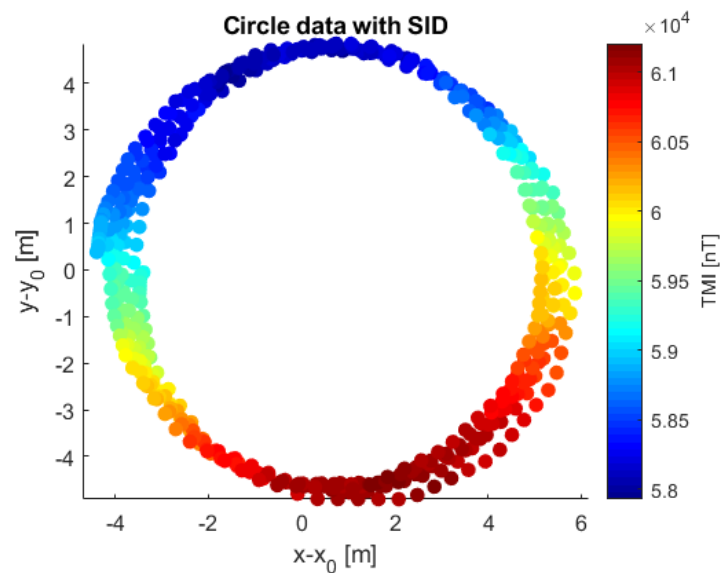
Hellisheiði Hi-Res magnetics



Reykjavik

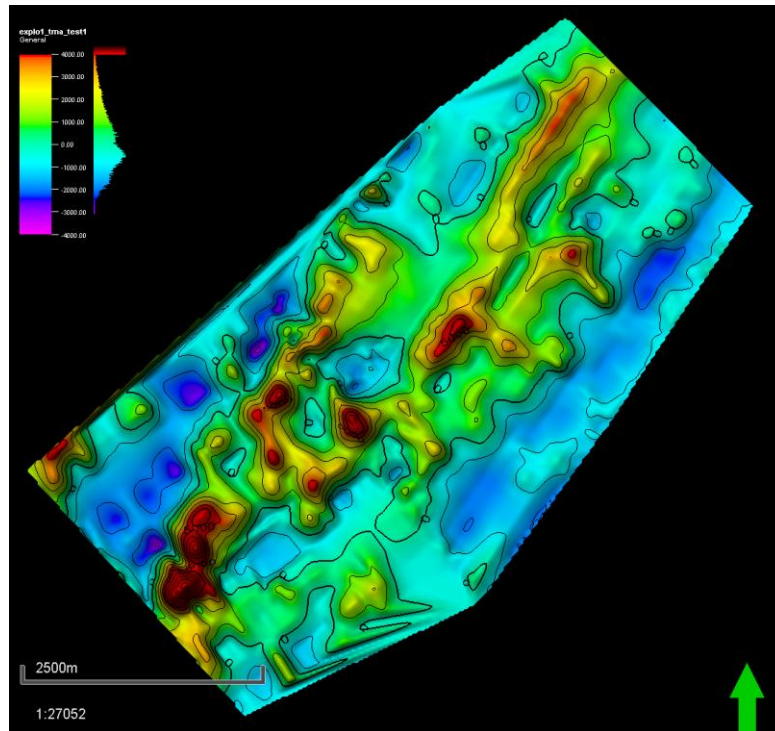
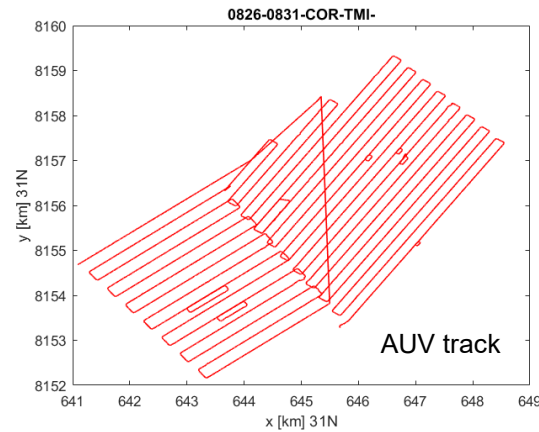


Selfoss

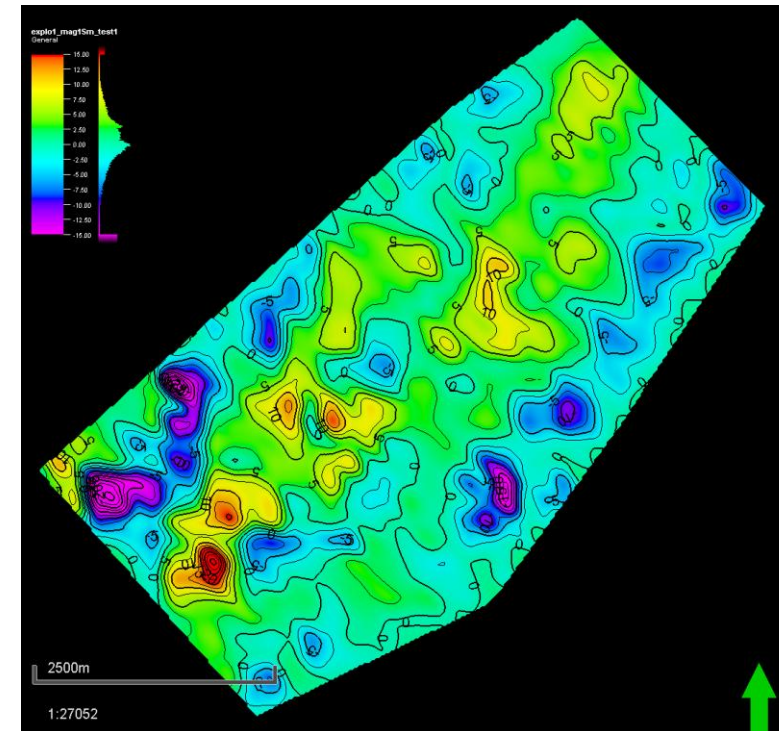


Magnetic inversion

- Data from NTNU MarMine 2016 cruise to Mohn's Ridge (Lim et al., 2019)
- Inhouse magnetic inversion code
- Remanent and induced magnetization

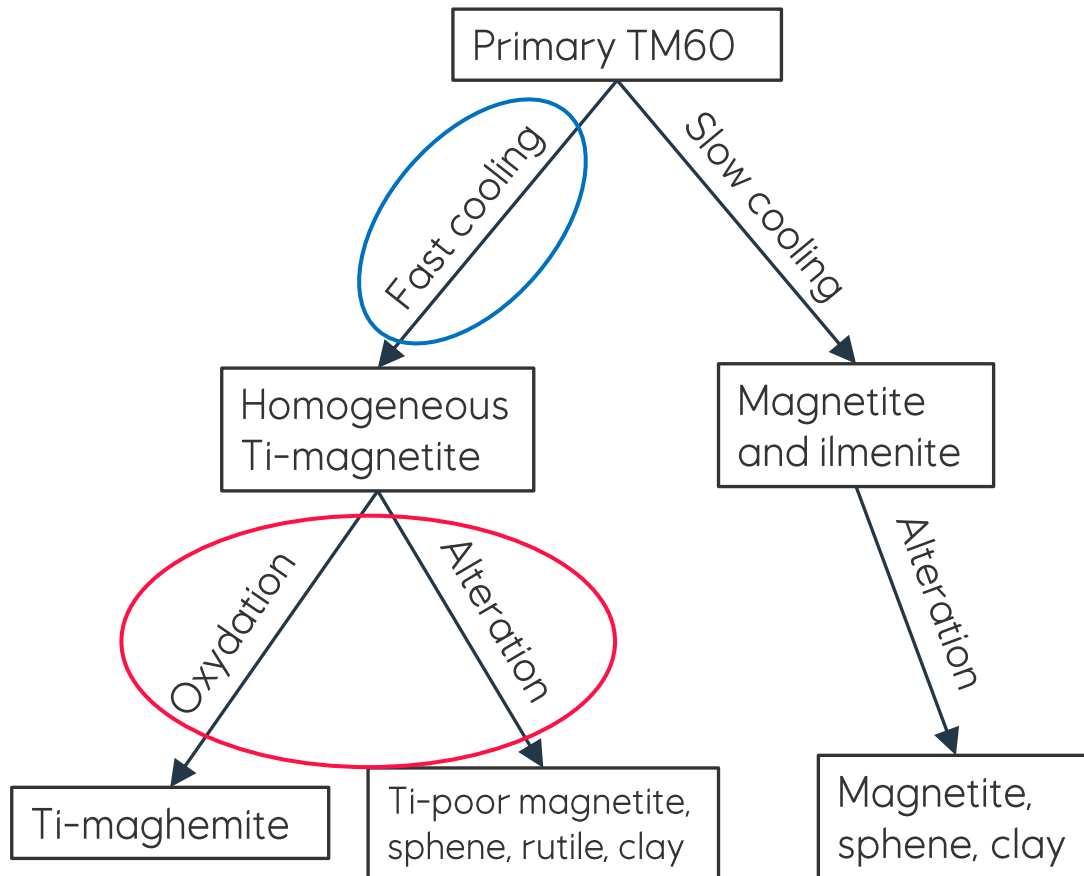


Total magnetic anomaly (TMA)

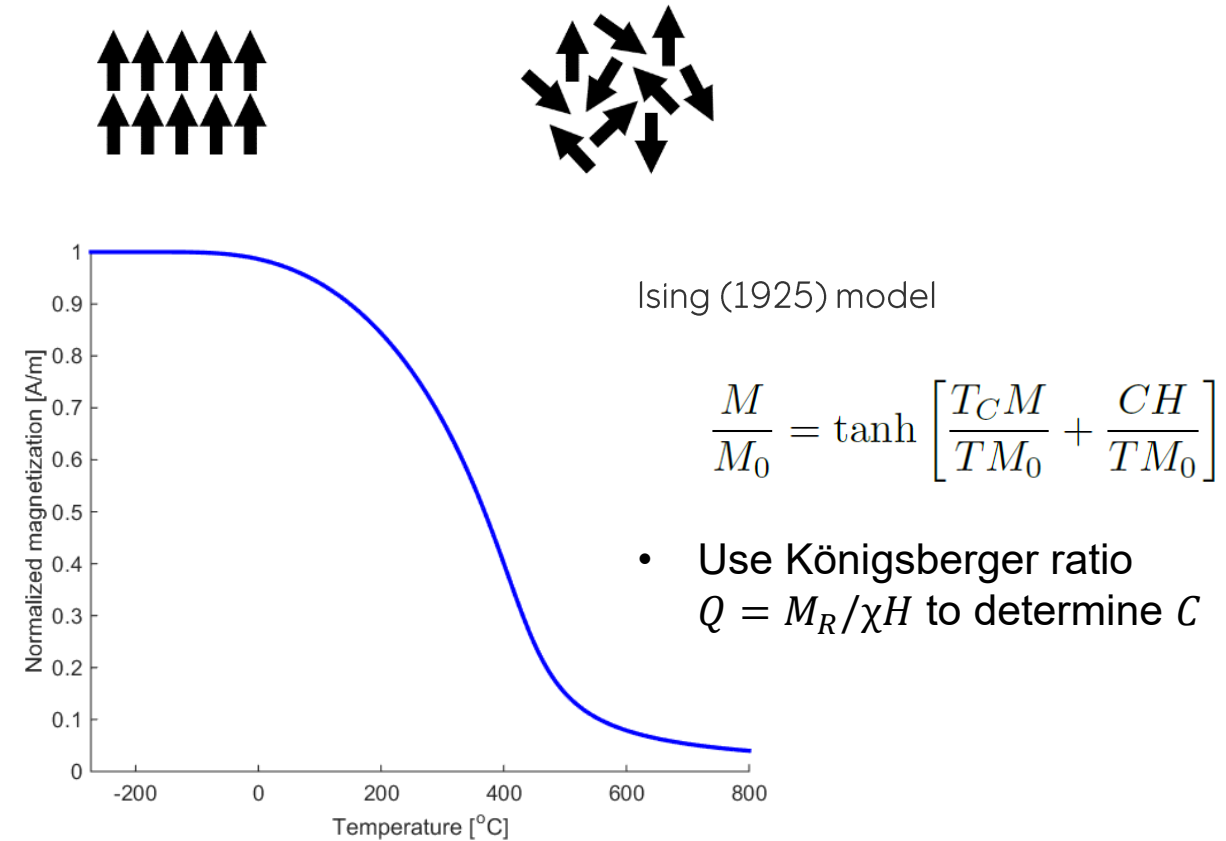


Magnetization from inversion

Magnetization



Adapted and simplified from Pariso and Johnson (1990)
Alteration in Costa Rica Ridge ODP data



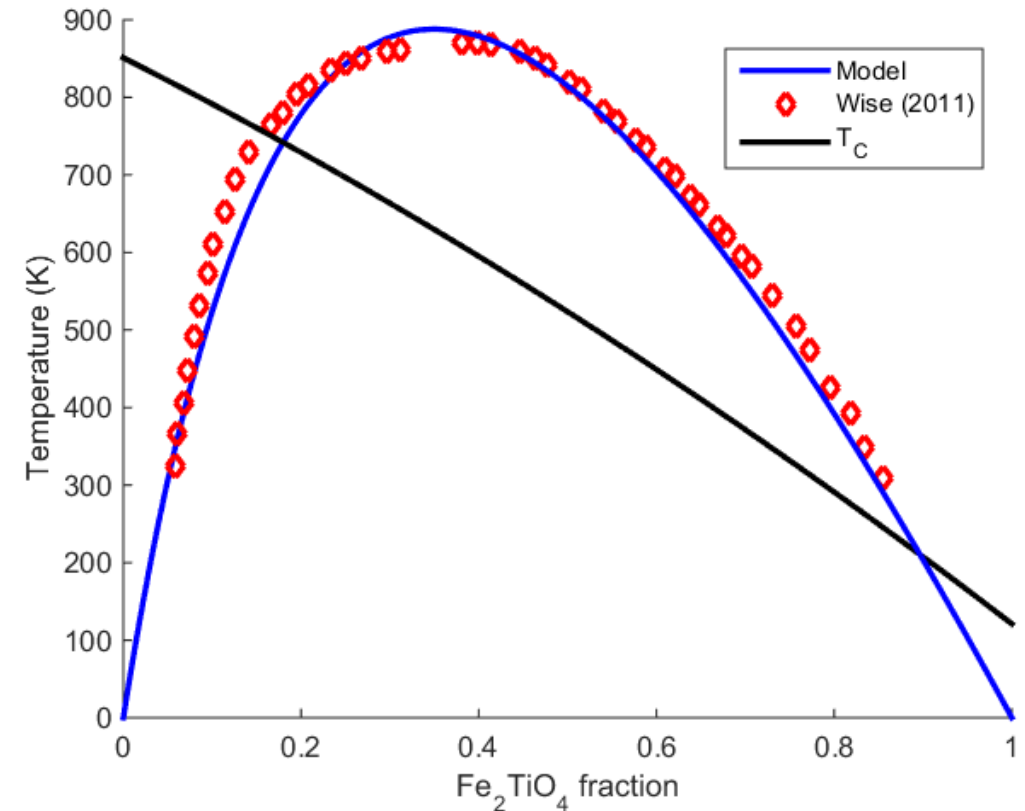
Hokstad, K., Alasonati Tašárová, Z., Sæther, B., and Tänävsuu-Milkeviciene, K., 2021, Inversion of Magnetic Data for Subsurface Temperature. World Geothermal Congress 2020+1

Magnetic mineral alteration

Factors that influence magnetic properties

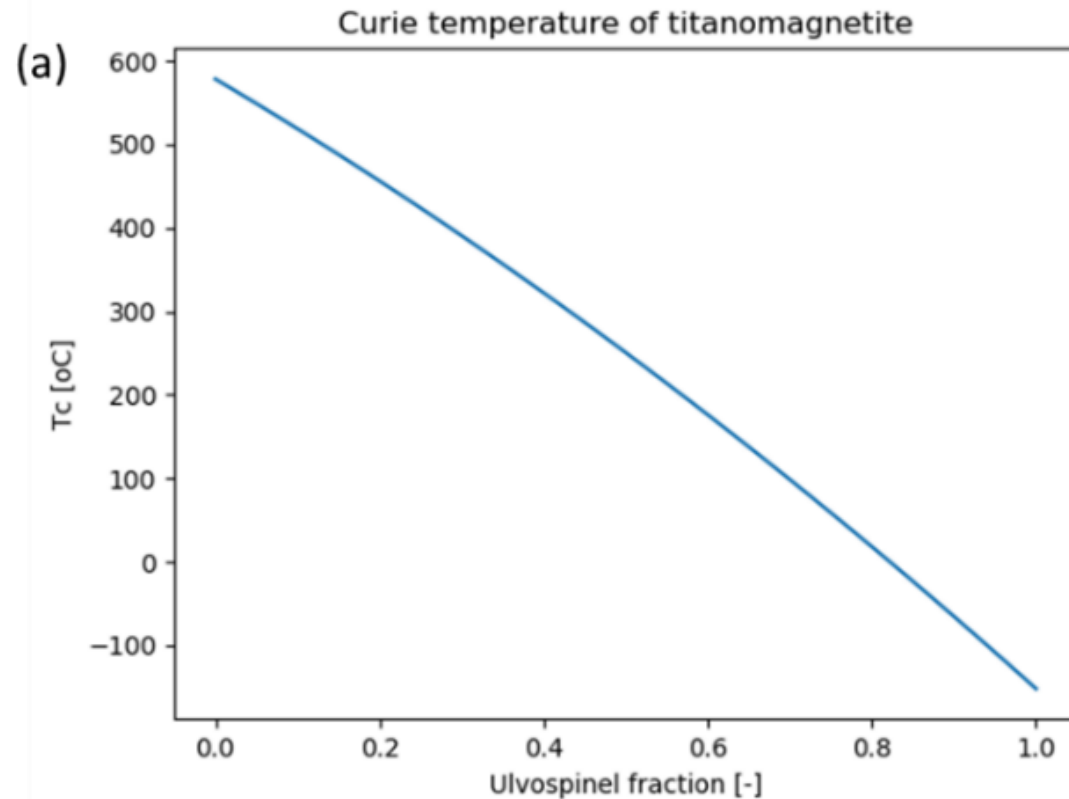
- High-temp oxidation
- Low-temp oxidation
- Solid exsolution
- Oxygen fugacity

Magnetic alteration impacts the Curie temperatures

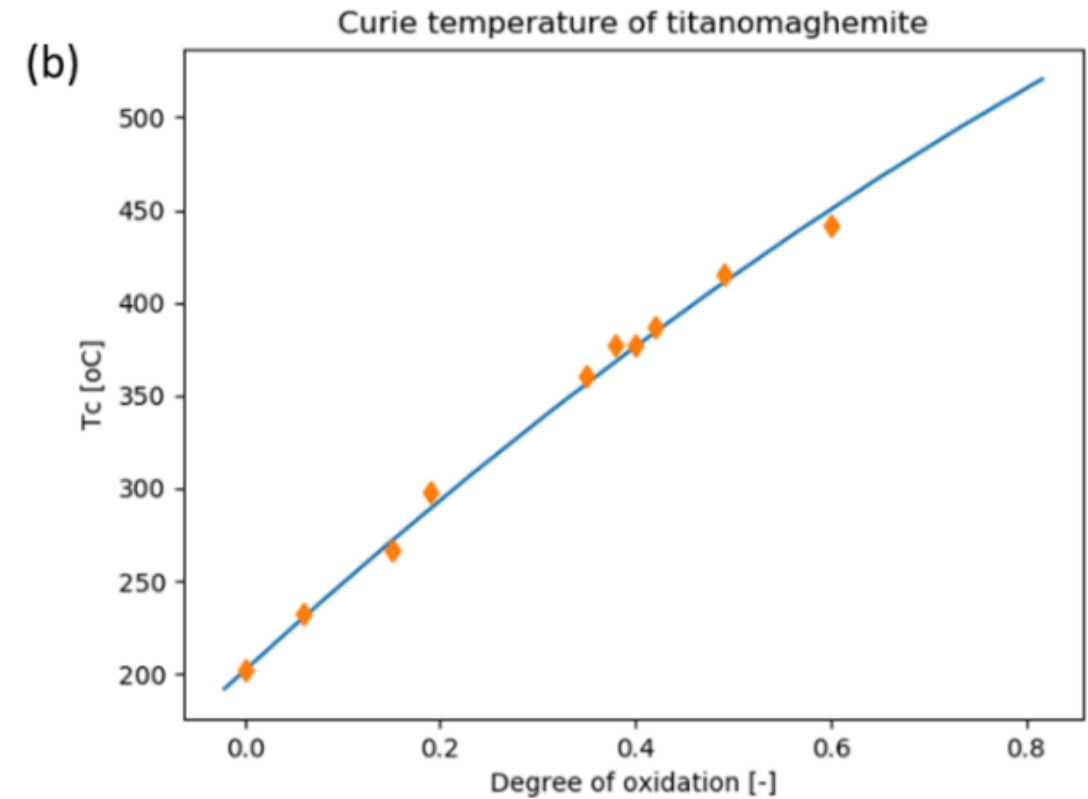


Solid exsolution of titanomagnetite

Curie temperature – the magnetic fingerprint

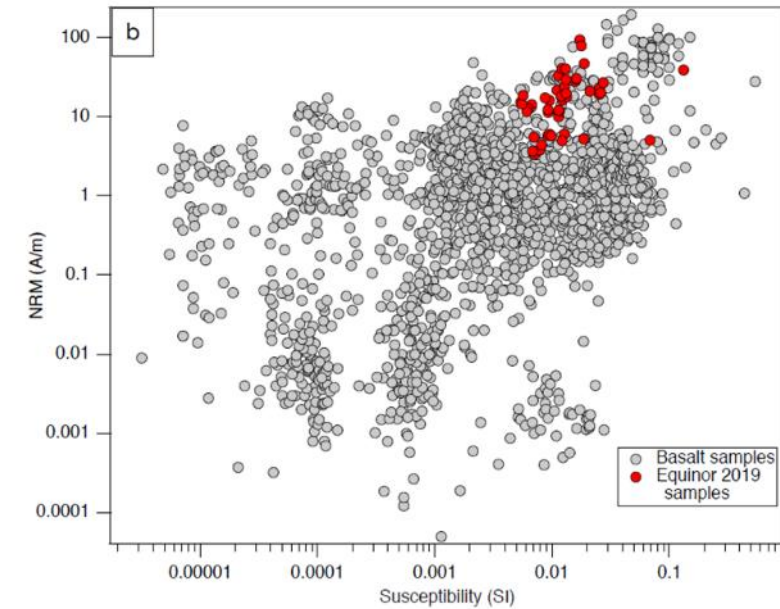
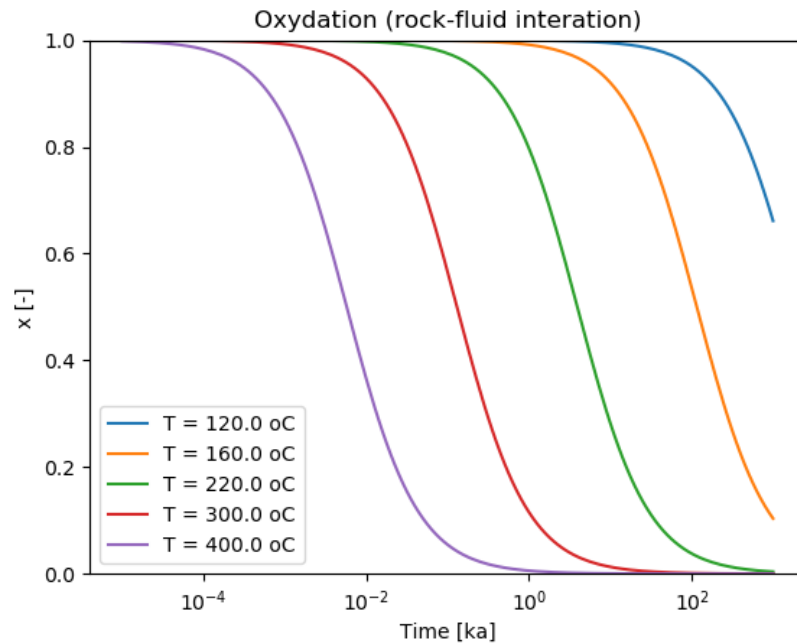


Lattard et al. (2006)



Data from Moskowitz and Banerjee (1981)

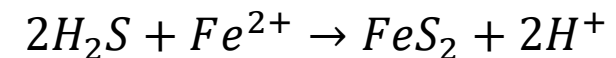
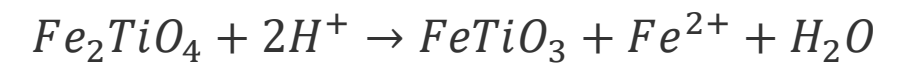
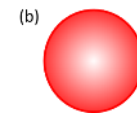
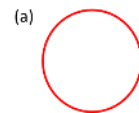
Demagnetization by rock-fluid interaction



Kinetic model: Mixed 1st/2nd order Arrhenius

$$\frac{dx}{dt} = -\left(\frac{k_1 - k_2}{x_0}\right)x^2 - k_2x$$

$$k_i = Ae^{-\frac{E_i}{RT}}$$



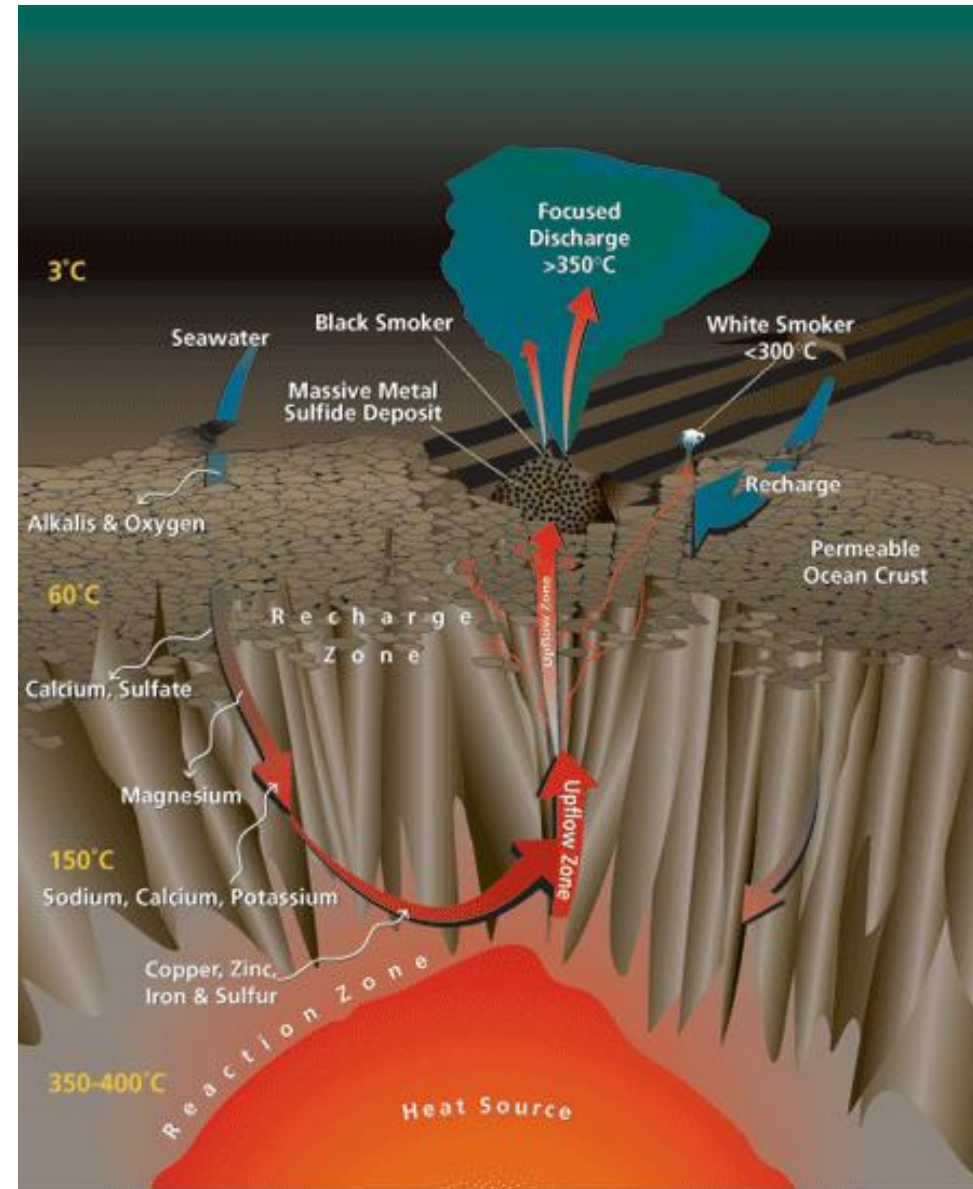
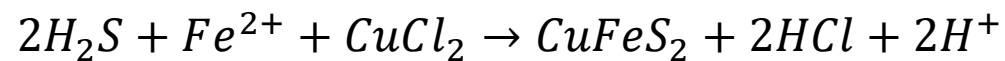
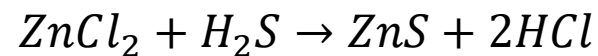
Reactive transport

Geothermal:

- Harvest energy from **active** hydrothermal systems

Seabed massive sulfides::

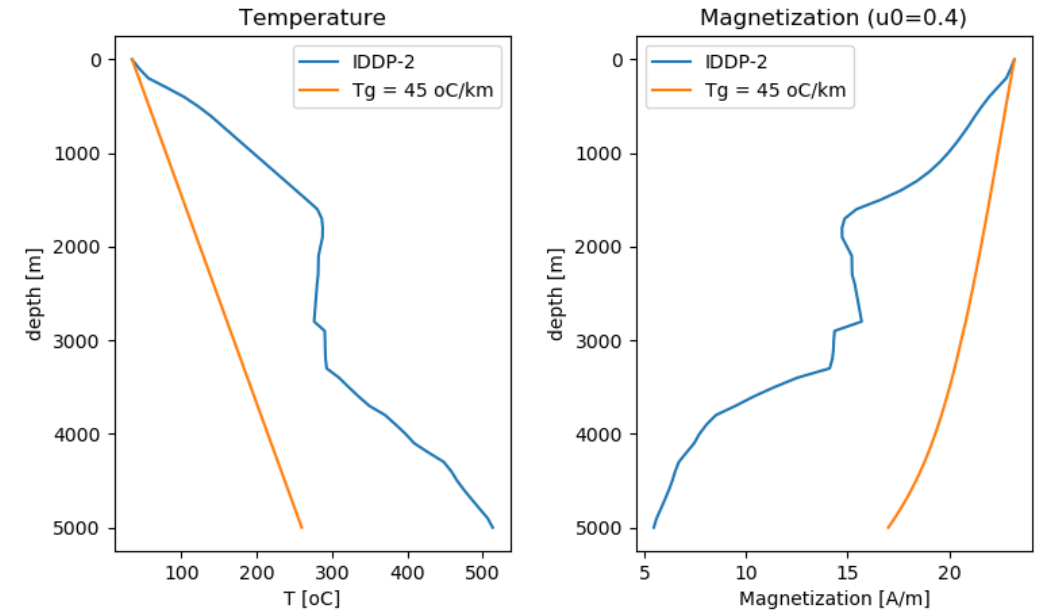
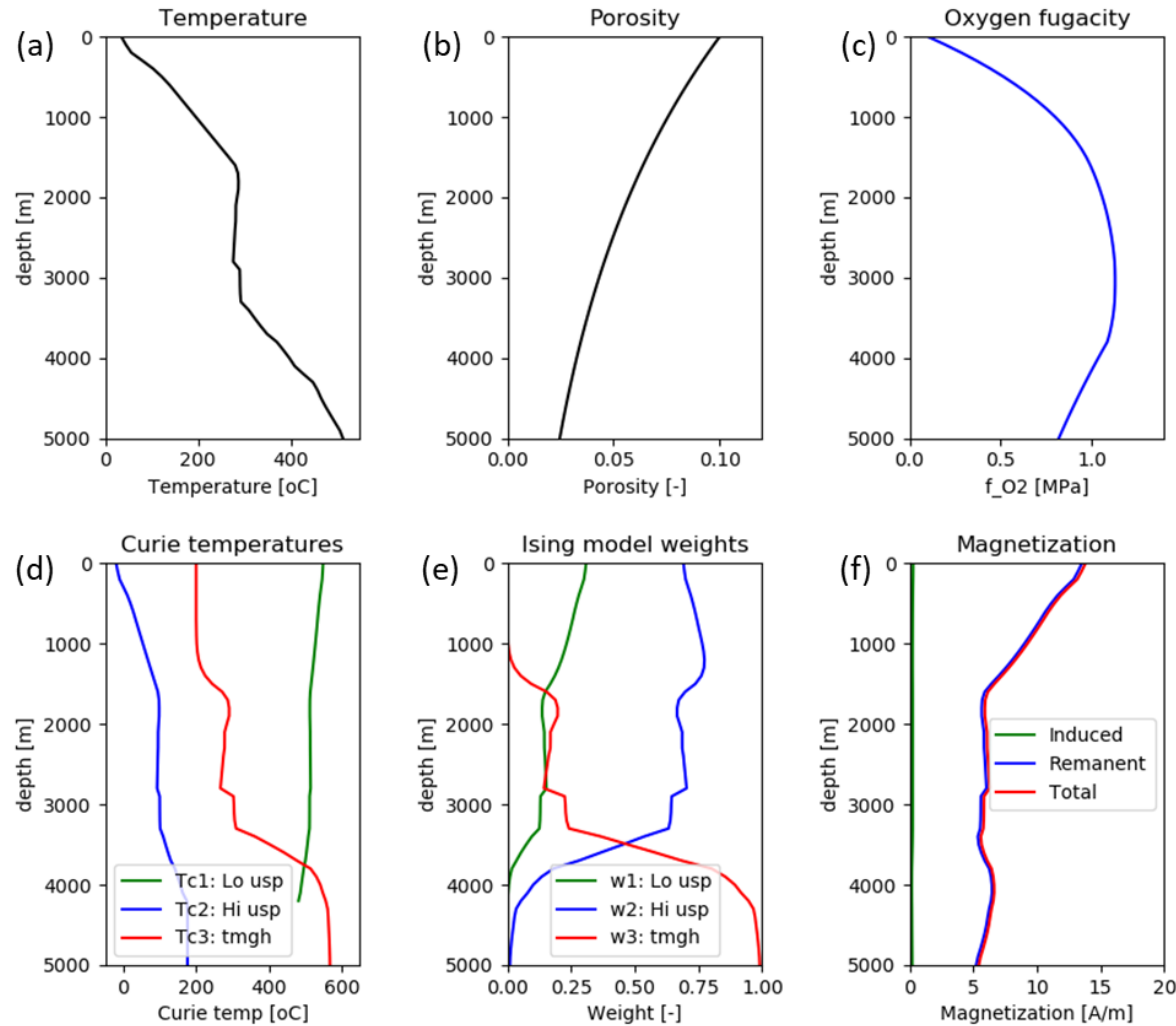
- Harvest minerals from **extinct** hydrothermal systems



«Basin modeling»
for hydrothermal
systems

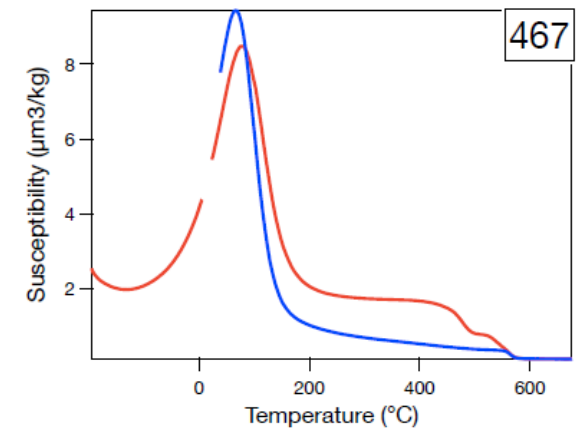
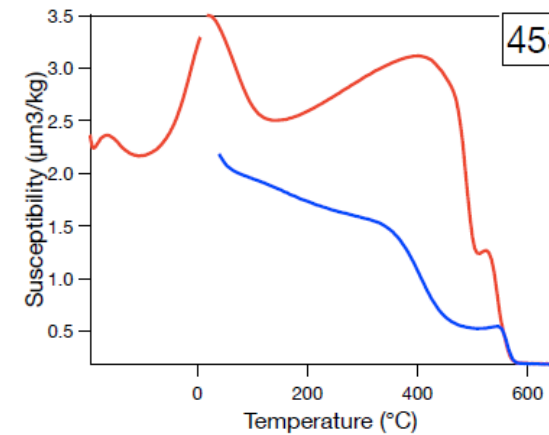
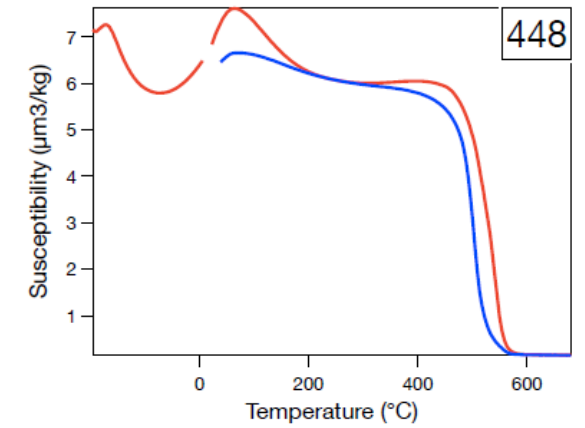
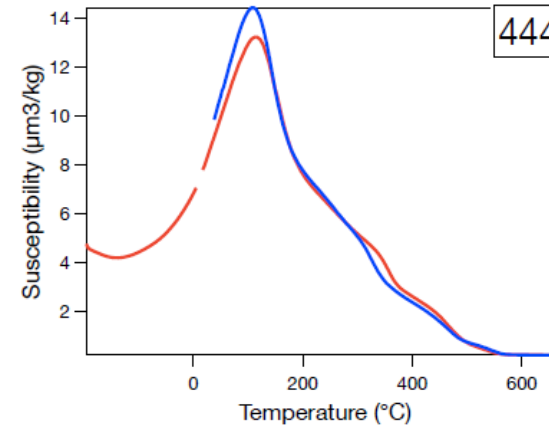
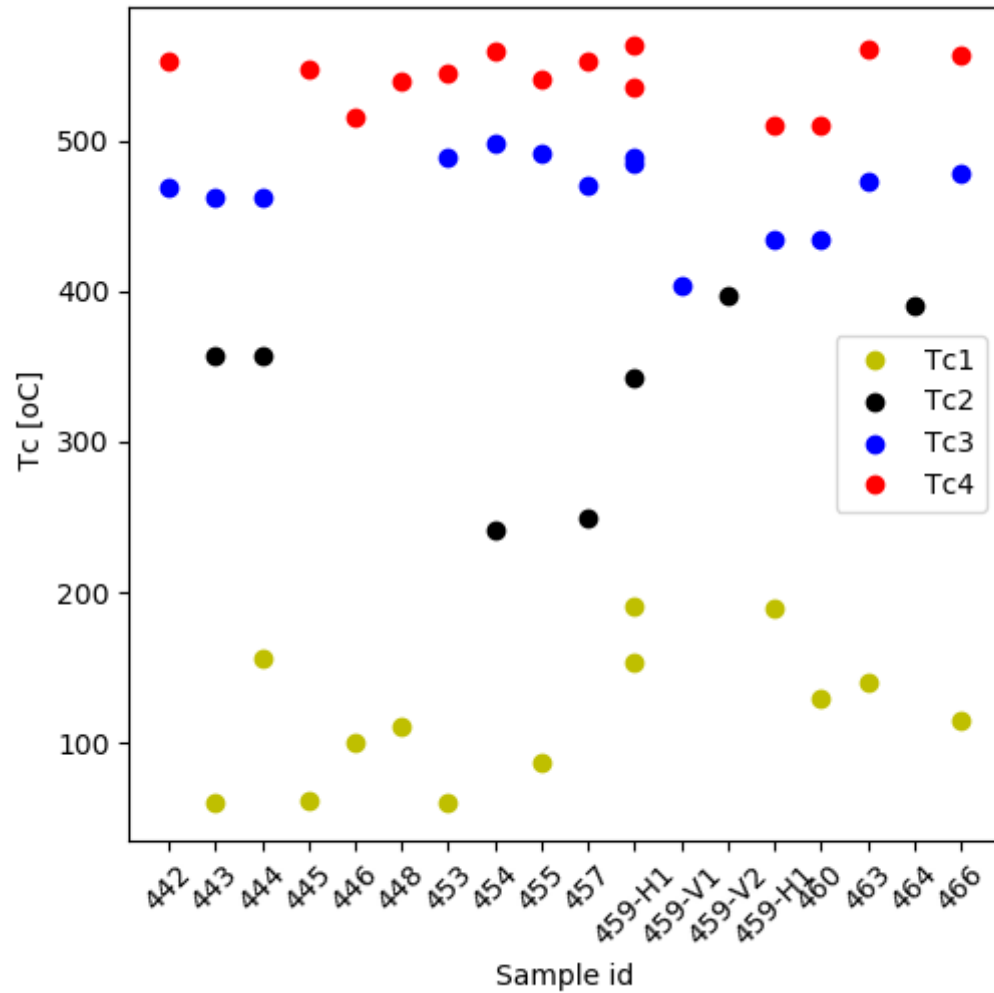
Source: Humphris (1998)

Forward modeling magnetization



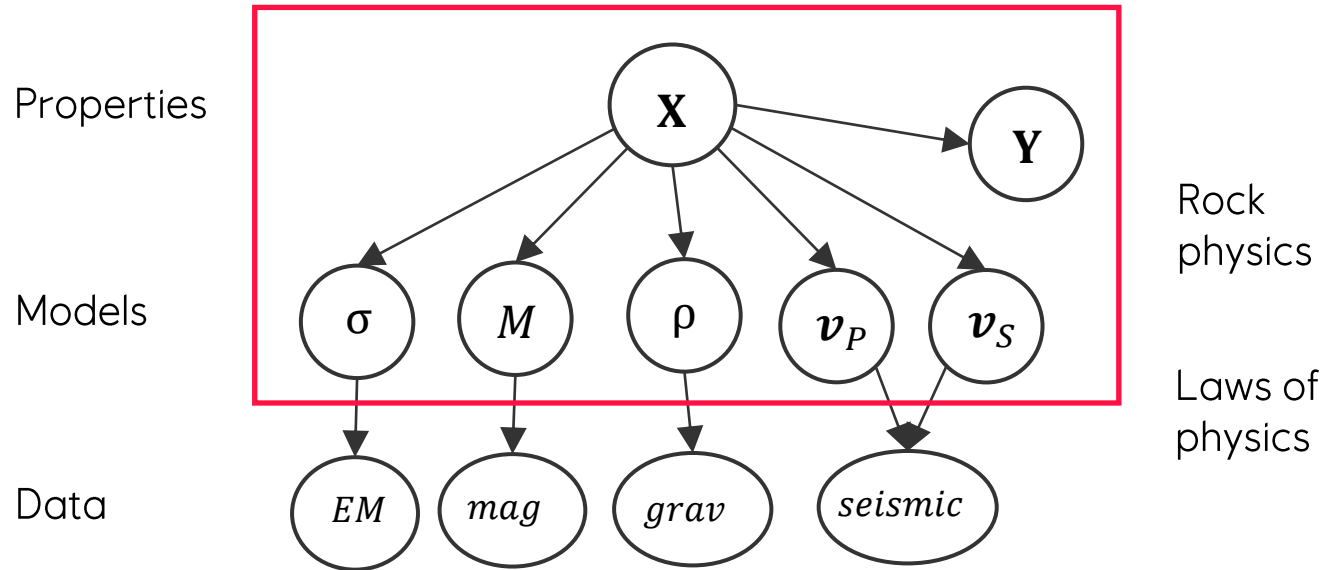
Forward modeling using temperature trend from IDDP-2 (multigeophysical inversion)

Hellisheiði Curie temperatures



Lab measurements by ter Maat and Church, NTNU (2020)

Multigeophysical inversion



Applications:

1. Basin modeling (heat flow)
2. Lithology and fluid prediction
- 3. Geothermal exploration**
- 4. Deep-sea mineral exploration**

Implementations:

- Inversion
- ML

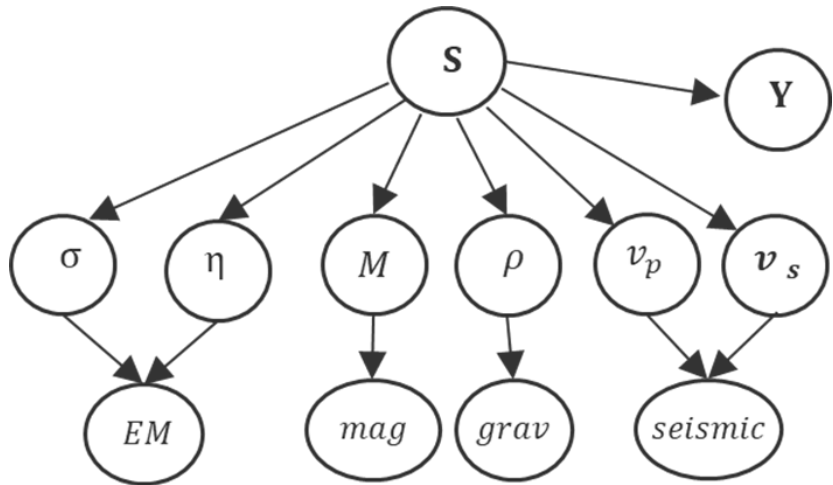


Thomas Bayes
(1701 - 1761)

Bayesian network for MGI

$$p(\mathbf{X}|m_1, \dots, m_n) = C \prod_{i=1}^n p(m_i|\mathbf{X})p(\mathbf{X}; \lambda)$$

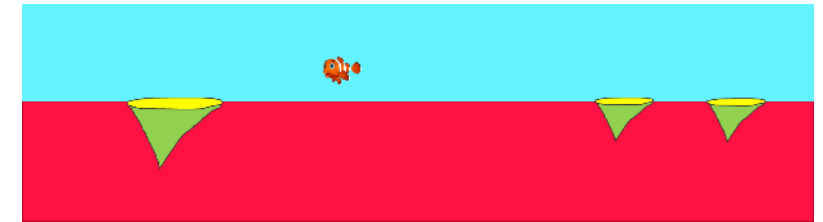
SMS exploration - Multigeophysical inversion (MGI)



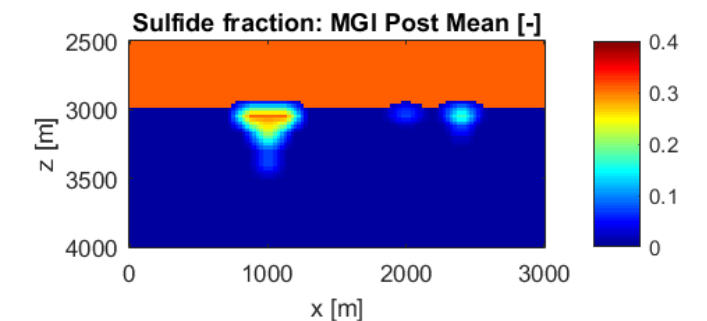
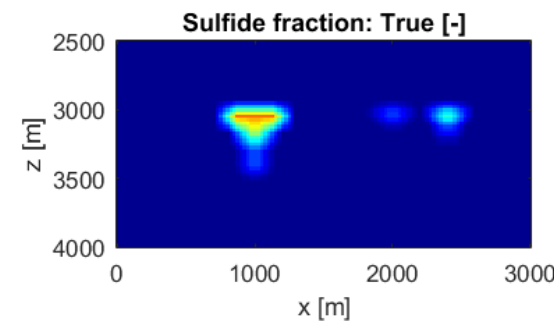
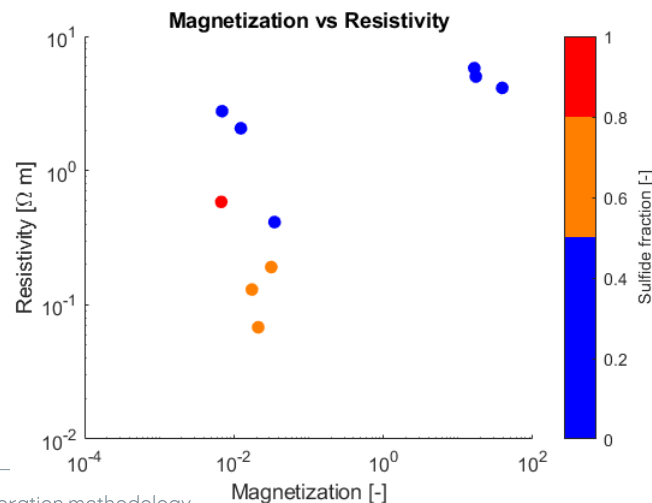
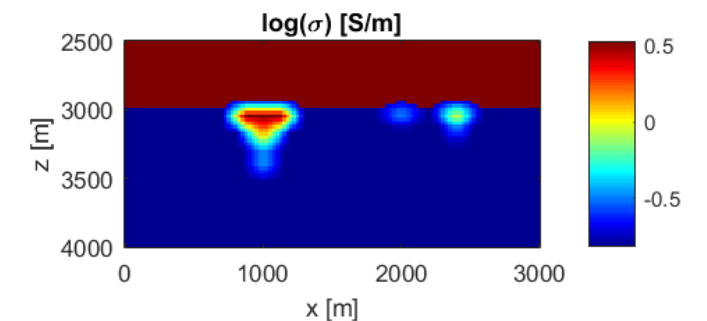
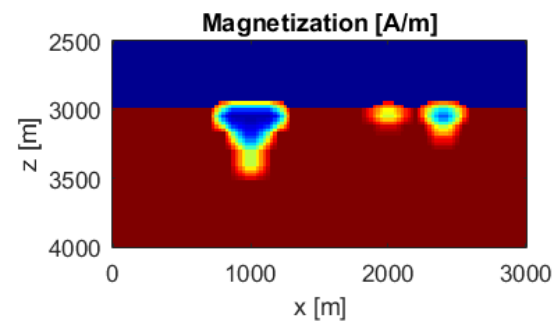
Multigeophysical inversion for metal sulfides

Geophysical data:

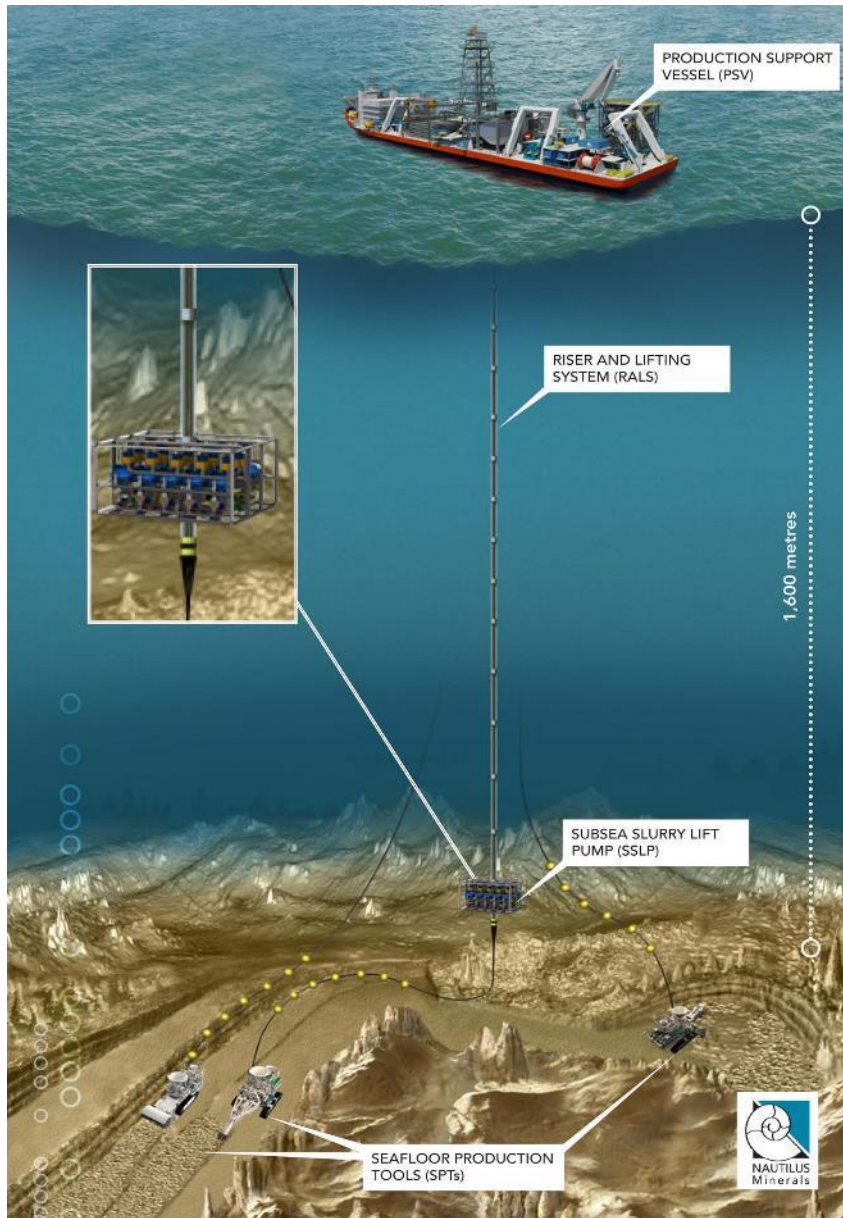
- Need to have?
- Nice to have?



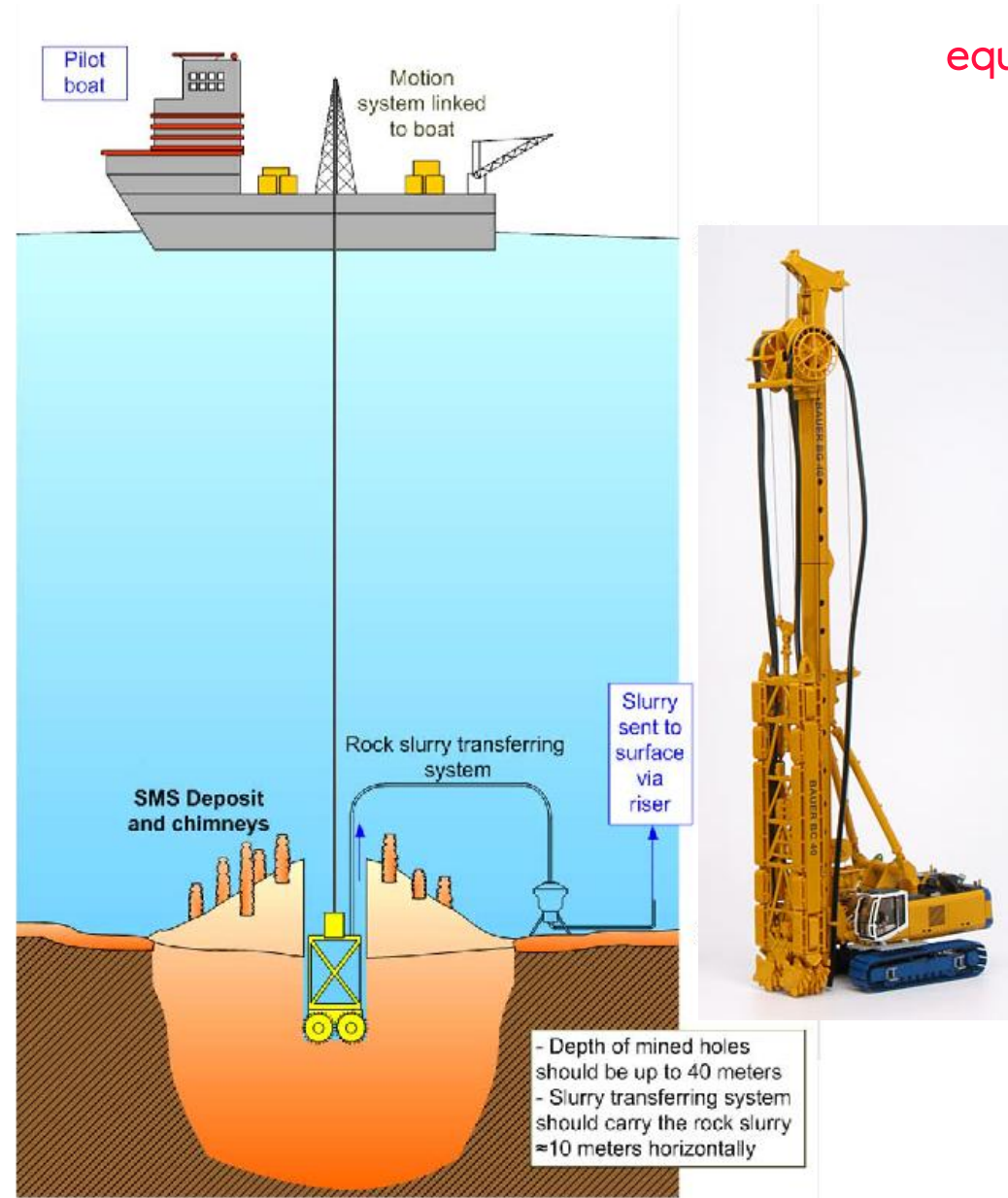
Synthetic test model



Hokstad, K., 2020, Method of estimating a mineral content of a geological structure. Pat. Appl. PCT/NO2020/050101.



Horizontal mining (Nautilus)



Vertical mining (Bauer, from Spagnoli 2016)

Environment

Active vents:

- Hosting a rich fauna
- Many endemic species
- Connectivity between vents and mounds?

Extinct vents:

- Background fauna or unique fauna?

Our «License to Operate»

- Don't touch the active vents
- Only extinct systems may, eventually, be exploited

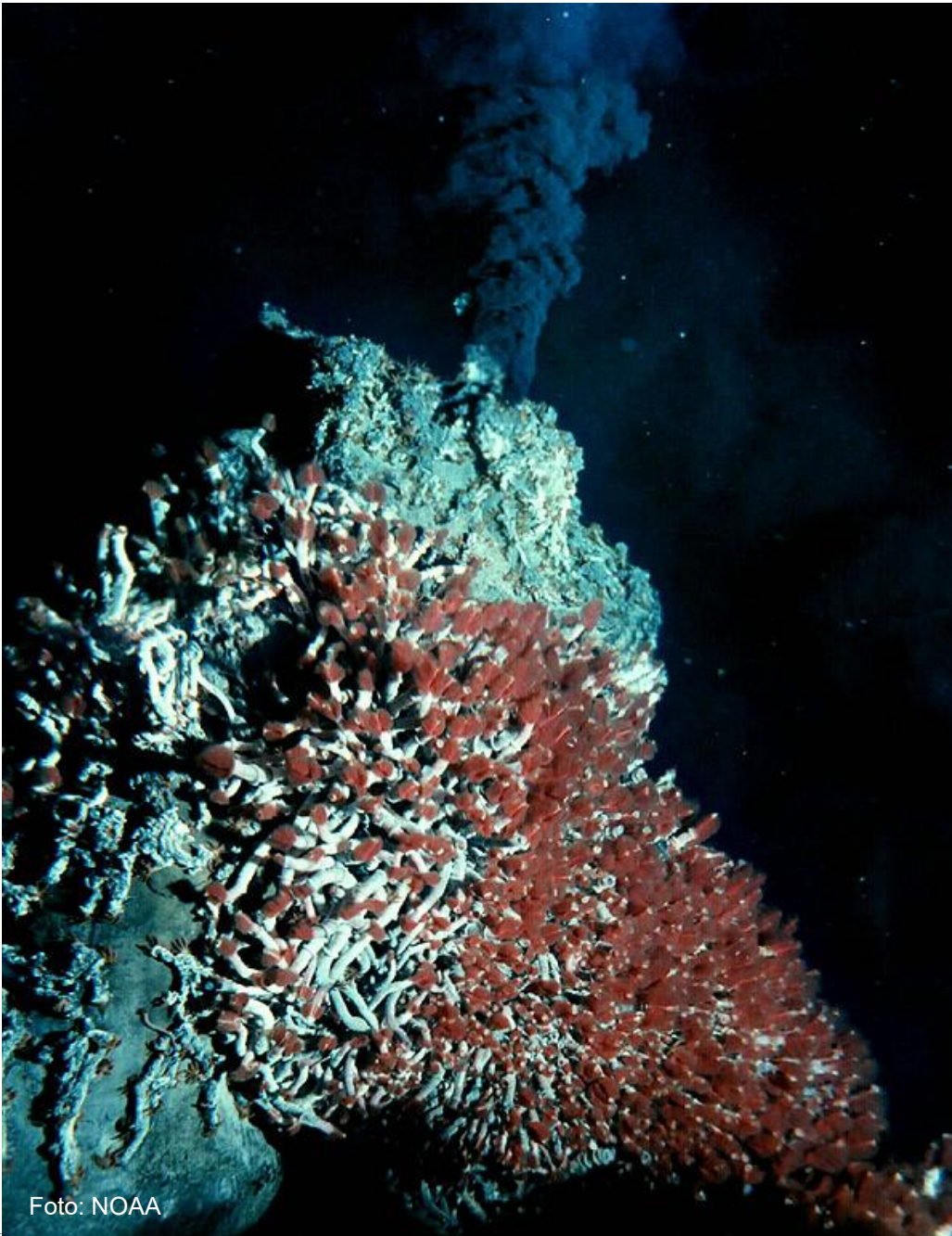


Foto: NOAA

Concluding remarks

- SMS exploration is a multidisciplinary challenge
- Two different scales
 - Regional: the main challenge
 - Local: a lot of work done in academia
- Geophysical data:
 - Need to have?
 - Nice to have?

Norwegian "Seabed Minerals Act"
adopted into law 1. July 2019

Why we **should** mine deep-sea minerals:

- Increased need for metals in the green shift
- Geopolitical considerations

Why we **should not** mine deep-sea minerals

- Protect the deep-sea environment and biology
- «No need for deep-sea minerals»

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Deep-Sea Minerals New Challenges for Geophysics

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